# MDD303 Probability

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# CHAPTER 1

# THE PROBABILISTIC MODEL

We consider a random space. We denote $\Omega$ the set of possible outcomes: it is an arbitrary set a priori.

## 1.1 Events

An event associated with a random experiment is a set of outcomes whose probability can be calculated. Let us denote $\mathcal{F}$ the collection of all events. Thus, $\mathcal{F} \subset \mathcal{P}(\Omega)$, the set of subsets of $\Omega$. In fact, we require that $\mathcal{F}$ be a tribe or $\sigma$-algebra over $\Omega$.

**Definition 1.1.1** (Tribe or $\sigma$-algebra)**.** A sigma-algebra or $\sigma$-algebra on $\Omega$ is a collection $\mathcal{F}$ of subsets of $\Omega$ that satisfies:

- $\Omega \in \mathcal{F}$
- If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$ (stability under complementation).
- If $(A_n)_{n\in\mathbb{N}}$ is a sequence of elements of $\mathcal{F}$, then their union $\bigcup_{n\in\mathbb{N}} A_n$ is also in $\mathcal{F}$ (stability under countable union).

**Remark 1.1.2.** A tribe necessarily contains $\emptyset$ and $\Omega$. Indeed, since $\mathcal{F}$ is non-empty, there exists $A \in \mathcal{F}$. By stability under complementation, $A^c \in \mathcal{F}$. By stability under union, $A \cup A^c = \Omega \in \mathcal{F}$. Finally, $\Omega^c = \emptyset \in \mathcal{F}$.

**Remark 1.1.3** (Reminder)**.** Two events $A$ and $B$ are said to be disjoint or incompatible if they cannot occur simultaneously, that is, if $A \cap B$ is the impossible event $\emptyset$.

## 1.2 Probability

Let $\Omega$ be a set equipped with a *sigma*-algebra $\mathcal{F}$.

**Definition 1.2.1** (Probability)**.** A probability on $(\Omega, \mathcal{F})$ is a function $P : \mathcal{F} \to [0, 1]$ that satisfies:

i) $P(\Omega) = 1$.

ii) If $(A_n)_{n\in\mathbb{N}}$ is a sequence of pairwise disjoint elements of $\mathcal{F}$, then

$$P\left(\bigcup_{n\in\mathbb{N}} A_n\right) = \sum_{n\in\mathbb{N}} P(A_n).$$

Let's consider the case where $\Omega$ is finite. In this case, we always take $\mathcal{F} = \mathcal{P}(\Omega)$. Let $A, B$ be two disjoint events. Applying ii) to the sequence $A_0 = A, A_1 = B, A_2 = A_3 = \cdots = \emptyset$, we obtain the finite additivity property:

ii') $\forall A, B$ disjoint, $P(A \cup B) = P(A) + P(B)$.

The modern theory of probabilities is built with axiom ii), stronger than ii'), which does not clash with intuition. But above all, it is a fundamental condition that enables proving numerous limit theorems.

**Definition 1.2.2** (Probability Space)**.** A probability space is a triplet $(\Omega, \mathcal{F}, P)$ where:

- $\Omega$ is a set;
- $\mathcal{F}$ is a *sigma*-algebra on $\Omega$;
- $P$ is a probability measure defined on $\mathcal{F}$.

The elements of $\mathcal{F}$ are called events. (Kolmogorov Axiomatics 1930).

## 1.3 Borel tribe

Let $\Omega$ be a set. If $\mathcal{F}_1, \mathcal{G}$ are two *sigma*-algebras on $\Omega$, we define their intersection as $\mathcal{F} \cap \mathcal{G} = \{A \subset \Omega; A \in \mathcal{F} \text{ and } A \in \mathcal{G}\}$. This is still a *sigma*-algebra.

More generally, if $(\mathcal{F}_i)_{i\in I}$ is a family of *sigma*-algebras on $\Omega$ indexed by the set $I$ whatever, then $\bigcap_{i\in I} \mathcal{F}_i = \{A \subset \Omega; \forall i \in I, A \in \mathcal{F}_i\}$ is still a *sigma*-algebra.

**Proposition 1.3.1** (Generated $\sigma$-algebra)**.** Let $\mathcal{A}$ be a collection of subsets of $\Omega$. There exists a unique *sigma*-algebra $\sigma(\mathcal{A})$ that satisfies:

i) $\mathcal{A} \subset \sigma(\mathcal{A})$.

ii) $\forall \mathcal{F}$ a *sigma*-algebra, $\mathcal{A} \subset \mathcal{F} \implies \sigma(\mathcal{A}) \subset \mathcal{F}$.

The *sigma*-algebra $\sigma(\mathcal{A})$ is called the *sigma*-algebra generated by $\mathcal{A}$; it is the smallest *sigma*-algebra containing $\mathcal{A}$.

**Proof.** Consider the following *sigma*-algebra: $\bigcap_{\mathcal{F}' \text{ tribu}, \mathcal{A}\subset\mathcal{F}'} \mathcal{F}'$. It is an intersection of *sigma*-algebras, so it is a *sigma*-algebra. By construction, it contains $\mathcal{A}$. It is contained in every *sigma*-algebra that contains $\mathcal{A}$. Thus, it is the smallest *sigma*-algebra that contains $\mathcal{A}$. □

**Remark 1.3.2.** This is the process used to define the subgroup generated by a family of elements (subspace generated...).

**Definition 1.3.3** (Borel $\sigma$-algebra $\mathcal{B}(\mathbb{R})$)**.** The Borel sigma-algebra on $\mathbb{R}$, denoted by $\mathcal{B}(\mathbb{R})$, is the smallest sigma-algebra containing the intervals. It is the sigma-algebra generated by one of the following families:

- Closed intervals $[a, b]$, with $a < b \in \mathbb{R}$.
- Open intervals $]a, b[$, with $a < b \in \mathbb{R}$.

- Half-infinite intervals $]-\infty, \lambda]$, with $\lambda \in \mathbb{R}$.
- Half-infinite intervals $[\lambda, +\infty[$, with $\lambda \in \mathbb{R}$.

## 1.4 Random variables

Random variables are the nymphs of probability spaces. A random variable (r.v.) is a function of the random experiment, so it is a priori a mapping $X : \Omega \to \mathbb{R}$. We want to be able to calculate the probability that a random variable falls within an arbitrary interval of $\mathbb{R}$. A necessary and sufficient condition for this is to require that $X$ be measurable.

**Definition 1.4.1** (Random Variable)**.** Let $(\Omega, \mathcal{F}, P)$ be a probability space. A random variable $X$ on $(\Omega, \mathcal{F}, P)$ is a measurable function $X : \Omega \to \mathbb{R}$ from $(\Omega, \mathcal{F})$ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, i.e.,

$$\forall B \in \mathcal{B}(\mathbb{R}), \quad X^{-1}(B) = \{\omega \in \Omega; X(\omega) \in B\} \in \mathcal{F}.$$

Thus, if $B \in \mathcal{B}(\mathbb{R})$, then $X^{-1}(B)$ is an event. It is simply denoted as $\{X \in B\}$ or even $X \in B$. In general, we strive to simplify formulas by omitting the variable $\omega$ whenever it is not essential.

**Convention:** Random variables are denoted by uppercase letters $X, Y, Z, U, V, W$, and their corresponding values by lowercase letters $x, y, z, u, v, w$. More generally, one can consider random variables taking values in $\mathbb{C}$, in $\mathbb{R}^n$, and indeed in any set equipped with a sigma-algebra.

**Definition 1.4.2** (Borel Function)**.** A function $f : \mathbb{R} \to \mathbb{R}$ is called Borel if it is measurable from $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, i.e., if $\forall B \in \mathcal{B}(\mathbb{R}), f^{-1}(B) = \{x \in \mathbb{R} : f(x) \in B\} \in \mathcal{B}(\mathbb{R})$.

**Example 1.4.3.** If $X$ is a random variable and $f$ is a Borel function, then $f(X)$ is still a random variable.

**Operations on random variables:** If $X, Y$ are random variables and $\alpha \in \mathbb{R}$, then $\alpha X$, $X + Y$, $XY$ are random variables. If $(X_n)$ is a sequence of random variables, then $\liminf X_n$, $\limsup X_n$ and $\lim X_n$ (if it exists) are random variables.

**Link with measure theory:** A space $\mathcal{E}$ equipped with a sigma-algebra $\mathcal{E}$ is called a measurable space. If $(E, \mathcal{E})$ and $(F, \mathcal{F})$ are two measurable spaces, a function $f : E \to F$ is measurable if $\forall A \in \mathcal{F}, f^{-1}(A) \in \mathcal{E}$. A probability $P$ is a (positive) measure with total mass 1.

## 1.5 The law of a random variable

Let $(\Omega, \mathcal{F}, P)$ be a probability space and let $X$ be a random variable defined on $\Omega$ taking values in $\mathbb{R}$.

**Definition 1.5.1** (Distribution of a Random Variable)**.** The law of the r.v. $X$, denoted as $P_X$, is the probability measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ defined by:

$$\forall B \in \mathcal{B}(\mathbb{R}) \quad P_X(B) = P(X \in B)(= P(X^{-1}(B)) = P(\{\omega \in \Omega; X(\omega) \in B\})).$$

Let us emphasize that the law of a random variable is a probability defined on the space of the values of the random variable, in our case, $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. We say that two random variables $X$ and $Y$ have the same law if $P_X = P_Y$. The primary object of interest when studying a random variable is its law. Indeed, we want to evaluate the probabilities of events associated with the random variable. "The law is fundamental!"

## 1.6 Expectation

**Definition 1.6.1** (Step Random Variable)**.** A random variable $\delta : \Omega \to \mathbb{R}$ is called 'stepped' if it takes a finite number of values. It can therefore be written as:

$$\delta = \sum_{i=1}^{n} \alpha_i \mathbb{I}_{A_i} \text{ where } \alpha_i \in \mathbb{R}, A_i \in \mathcal{F}.$$

Its expectation is its integral with respect to $P$, given by $E(\delta) = \int \delta dP = \sum_{i=1}^{n} \alpha_i P(A_i)$.

**Definition 1.6.2** (Expected Value of a Positive R.V.)**.** If $X : \Omega \to [0, +\infty]$ is a non-negative or positive random variable, possibly taking the value $+\infty$, its expectation is defined as:

$$E(X) = \sup\{E(\delta), \delta \text{ step}, 0 \leq \delta \leq X\}.$$

This is also the integral of $X$ with respect to $P$. We denote $E(X) = \int X dP$. Note that we may have $E(X) = +\infty$.

**Definition 1.6.3** (Expected Value of an Arbitrary R.V.)**.** If $X : \Omega \to \mathbb{R}$ is of any sign, we write $X = X^+ - X^-$ where:

- $X^+ = \max(X, 0)$ is the positive part.
- $X^- = -\min(X, 0)$ is the negative part.

$X^+$ and $X^-$ are two positive random variables. When $E(X^+) < \infty$ and $E(X^-) < \infty$, we set:

$$E(X) = \int X dP = E(X^+) - E(X^-).$$

Since moreover, $|X| = X^+ + X^-$, we have the equivalence: $E(X^+) < \infty, E(X^-) < \infty \iff E(|X|) < \infty$. In this case, its expectation is $E(X) = \int X dP$. We denote $\mathcal{L}^1(\Omega, \mathcal{F}, P)$ the space of integrable random variables.

## 1.7 Variance

**Definition 1.7.1** (Variance)**.** If $X$ is a random variable such that $E(X^2) < \infty$ (which implies that $X$ is integrable), its variance is defined as:

$$\begin{aligned} \mathrm{Var}(X) &= E[(X - E(X))^2] \\ &= E(X^2) - [E(X)]^2 \end{aligned}$$

# CHAPTER 2

# LEBESGUE MEASURE AND PROBABILITY LAWS

## 2.1 The Lebesgue measure

**Definition 2.1.1.** A measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is an application $\mu$ from $\mathcal{B}(\mathbb{R})$ to $[0, +\infty[$ such that $\mu(\emptyset) = 0$ and that is $\sigma$-additive, that is:

$$\forall (B_n)_{n\in\mathbb{N}} \in \mathcal{B}(\mathbb{R})^{\mathbb{N}}, \quad \left(\forall n \neq m, B_n \cap B_m = \emptyset \implies \mu\left(\bigcup_{n\in\mathbb{N}} B_n\right) = \sum_{n\in\mathbb{N}} \mu(B_n)\right)$$

**Theorem 2.1.2.** There exists a unique measure $\lambda$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ that satisfies:

$$\forall a, b \in \mathbb{R}, a < b, \quad \lambda([a, b]) = b - a.$$

This measure is called the **Lebesgue measure** and is denoted by $\lambda$.

**Remark 2.1.3.** The existence of such a measure is by no means obvious; it is a profound mathematical result.

### Proof idea

For any subset $A$ of $\mathbb{R}$, we define the Lebesgue outer measure $\lambda^*(A)$ by:

$$\lambda^*(A) = \inf\left\{\sum_{i\in I}(b_i - a_i); (a_i)_{i\in I}, (b_i)_{i\in I} \text{ families of real numbers such that } \forall i \in I, a_i < b_i, A \subset \bigcup_{i\in I} ]a_i, b_i[\right\}$$

$\lambda^*$ is not a measure, it is subadditive but not $\sigma$-additive. We show that the restriction of $\lambda^*$ to $\mathcal{B}(\mathbb{R})$ is $\sigma$-additive and we then take $\lambda = \lambda^*|_{\mathcal{B}(\mathbb{R})}$.

### First properties

- For all $x \in \mathbb{R}$, $\lambda(\{x\}) = 0$.
- If $A \subset \mathbb{R}$ is countable, then $\lambda(A) = \sum_{x\in A} \lambda(\{x\}) = 0$ by $\sigma$-additivity.
- There exist Borel sets $C$ in $\mathbb{R}$ that are uncountable such that $\lambda(C) = 0$.

## 2.2 Uniform law on $[0,1]$

The uniform law on $[0,1]$ is the probability measure $P$ on $([0,1],\mathcal{B}([0,1]))$ obtained as the restriction of the Lebesgue measure $\lambda$ to $[0,1]$:

$$\forall B \in \mathcal{B}([0,1]), \quad P(B) = \lambda(B).$$

We want to construct a mathematical model of the experiment consisting of drawing a number at random in $[0,1]$, uniformly. We take the measurable space $(\Omega,\mathcal{F}) = ([0,1],\mathcal{B}([0,1]))$. The Borel sigma-algebra $\mathcal{B}([0,1])$ is, at choice:

- the tribe generated by the sub-intervals of $[0,1]$,
- or the trace tribe of $\mathcal{B}(\mathbb{R})$ on $[0,1]$, that is, $\{B \cap [0,1] : B \in \mathcal{B}(\mathbb{R})\}$.

This space is equipped with the probability $P = \lambda|_{[0,1]}$. We then define a random variable $X$ by setting:

$$\begin{aligned} X : \Omega = [0,1] &\to \mathbb{R} \\ \omega &\mapsto X(\omega) = \omega \end{aligned}$$

It is said that $X$ is a number chosen randomly and uniformly in $[0,1]$. The law of $X$, denoted $P_X$, is $\lambda|_{[0,1]}$. Indeed, for all $B \in \mathcal{B}([0,1])$:

$$P_X(B) = P(X \in B) = P(\{\omega \in \Omega : X(\omega) \in B\}) = P(B) = \lambda(B).$$

## 2.3 Density Laws

**Definition 2.3.1.** A probability law $\mu$ on $\mathbb{R}$ is said to have a **density** if there exists a positive Borel function $f$ such that:

$$\forall A \in \mathcal{B}(\mathbb{R}), \quad \mu(A) = \int_A f d\lambda = \int_A f(x)dx.$$

The function $f$ is called the density (or a density) of $\mu$. A density is a positive Borel function that satisfies:

$$\int_{\mathbb{R}} f d\lambda = \int_{\mathbb{R}} f(x)dx = 1.$$

The density of a law is not unique; it is defined up to a set of Lebesgue measure zero. Conversely, given a positive Borel function $f$ satisfying this condition, one can construct a probability law on $\mathbb{R}$ associated with $f$ using the above formula.

**Example 2.3.2** (Fundamental Examples).

- **Uniform distribution** on an interval $[a,b]$, denoted $\mathcal{U}([a,b])$. It is the distribution with density $\frac{1}{b-a}\mathbb{1}_{[a,b]}(x)$.
- **Exponential distribution** with parameter $\alpha > 0$, denoted $\mathrm{Exp}(\alpha)$. It is the distribution with density $\alpha e^{-\alpha x}\mathbb{1}_{\mathbb{R}_+}(x)$.
- **Normal distribution** or Gaussian with parameters $m, \sigma^2$, denoted $\mathcal{N}(m,\sigma^2)$. It is the distribution with density $\frac{1}{\sqrt{2\pi}\sigma}\exp\left(-\frac{(x-m)^2}{2\sigma^2}\right)$.

## 2.4 Classification of laws on $\mathbb{R}$

A point $x \in \mathbb{R}$ is an **atom** of the law $\mu$ if $\mu(\{x\}) > 0$. We say that the law $\mu$ is **diffuse** if it has no atoms. Three "pure" types of laws are distinguished:

- **Discrete laws:** A measure $\mu$ is said to be discrete if there exists a finite or countable set $D$ such that $\mu(D) = 1$. We then have $\mu = \sum_{x\in D}\mu(\{x\})\delta_x$.

- **Laws with density:** A measure $\mu$ is said to have a density if there exists a Borel function $f \geq 0$ such that $\forall A \in \mathcal{B}(\mathbb{R})$, $\mu(A) = \int_A f d\lambda$.
- **Singular laws:** A measure $\mu$ is said to be singular if it is diffuse and if there exists a set $C$ such that $\mu(C) = 1$ and $\lambda(C) = 0$.

Every probability measure on $\mathbb{R}$ can be decomposed as a convex combination of these three types:

$$\mu = \alpha_d \mu_d + \alpha_a \mu_a + \alpha_s \mu_s$$

where $\mu_d, \mu_a, \mu_s$ are discrete, absolutely continuous, and singular, respectively, and $\alpha_d, \alpha_a, \alpha_s \in [0, 1]$ with $\alpha_d + \alpha_a + \alpha_s = 1$. This representation is unique.

## 2.5 The Lebesgue integral

Suppose we have a theory of integration. Given a function $f \geq 0$ such that $\int_{\mathbb{R}} f(x)dx = 1$, we can define a probability measure by $\mu(A) = \int_A f(x)dx$ for all sets $A$ allowed by the theory. For example, with the Riemann integral, we can only integrate "sufficiently regular" functions (e.g., continuous ones), and the probability is only defined on intervals or finite unions of intervals. The preferred choice for probability theory is the Lebesgue integral, which allows integration over all Borel sets.

### Summary of the construction of the Lebesgue integral

1. **Layered functions:** A Borel function $s$ is said to be layered if it takes a finite number of values. It has the form $s = \sum_{i=1}^n a_i \mathbb{1}_{A_i}$, where $a_i \in \mathbb{R}$ and $A_i \in \mathcal{B}(\mathbb{R})$. Its integral with respect to $\lambda$ is:

$$\int s \, d\lambda = \sum_{i=1}^n a_i \lambda(A_i).$$

2. **Positive functions:** If $f$ is a Borel function that is positive or zero, the integral of $f$ with respect to $\lambda$ is:

$$\int f \, d\lambda = \sup \left\{ \int s \, d\lambda \ ; \ s \text{ layered}, \ 0 \leq s \leq f \right\}.$$

   Note that the above supremum may be infinite.

3. **Functions of arbitrary sign:** Let $f$ be a Borel function. We say that $f$ is integrable if $\int |f| \, d\lambda < +\infty$. In this case, we decompose $f$ into its positive part $f^+ = \max(f, 0)$ and its negative part $f^- = \max(-f, 0) = -\min(f, 0)$. We then define:

$$\int f \, d\lambda = \int f^+ \, d\lambda - \int f^- \, d\lambda.$$

   We denote $L^1(\mathbb{R}, \lambda)$ the set of $\lambda$-integrable functions.

## 2.6 Transfer formula

When working with random variables, the first step involves performing manipulations in the probability space $(\Omega, \mathcal{F}, P)$. When dealing with the specific distributions of the random variables, the computation is transferred to $\mathbb{R}$.

**Proposition 2.6.1** (Transfer Formula)**.** Let $(\Omega, \mathcal{F}, P)$ be a probability space, $X$ a random variable taking

values in $\mathbb{R}$, and $g : \mathbb{R} \to \mathbb{R}$ a Borel function such that $g(X)$ is an integrable random variable. Then

$$E(g(X)) = \int_\Omega g(X(\omega))\, dP(\omega) = \int_{\mathbb{R}} g(x)\, dP_X(x).$$

**Proof** (Proof Idea). The proof proceeds in successive steps by showing that the formula holds for increasingly general classes of functions.

1. **Case of an indicator function:** Let us start with $g = \mathbb{1}_B$ where $B \in \mathcal{B}(\mathbb{R})$.

$$\begin{aligned}\int_\Omega \mathbb{1}_B(X(\omega))\, dP(\omega) &= P(\{\omega \in \Omega; X(\omega) \in B\}) \\ &= P(X \in B) = P_X(B) \\ &= \int_{\mathbb{R}} \mathbb{1}_B(x)\, dP_X(x).\end{aligned}$$

   The formula is thus verified for indicator functions.

2. **Case of a positive simple function:** By the linearity of the integral, the formula extends to positive simple functions $s = \sum a_i \mathbb{1}_{B_i}$.

3. **Case of a positive Borel function:** We use a monotone convergence argument. Let $\mathcal{L}$ be the collection of positive Borel functions for which the transfer formula holds. We show that $\mathcal{L}$ satisfies the following properties:

   (i) $\forall B \in \mathcal{B}(\mathbb{R}), \mathbb{1}_B \in \mathcal{L}$.

   (ii) $\forall f, g \in \mathcal{L}$, $\forall \alpha, \beta > 0$, $\alpha f + \beta g \in \mathcal{L}$ (stability under positive linear combination).

   (iii) $\mathcal{L}$ is stable under increasing limits: if $(f_n)$ is a sequence of functions in $\mathcal{L}$ such that $f_n \leq f_{n+1}$ and $f_n \to f$ pointwise, then $f \in \mathcal{L}$.

   By the monotone class lemma, $\mathcal{L}$ contains all positive Borel functions.

4. **General case:** For a function $g$ of any sign, we apply the result to its positive part $g^+$ and its negative part $g^-$.

□

### 2.6.1 Application to Expectation and Variance

Let $X$ be an integrable random variable on $(\Omega, \mathcal{F}, P)$ with values in $\mathbb{R}$.

**Expectation** The expectation of $X$ is defined by $E(X) = \int_\Omega X(\omega) dP(\omega)$. By applying the transfer formula with $g(x) = x$, we get:

$$E(X) = \int_{\mathbb{R}} x\, dP_X(x).$$

If the law of $X$, $P_X$, has a density $f$, then:

$$E(X) = \int_{x \in \mathbb{R}} x f(x) dx.$$

**Variance** If $X$ is square-integrable, its variance is defined by $\mathrm{Var}(X) = E[(X - E(X))^2]$. By setting $m = E(X)$ and applying the transfer formula to the function$g(x) = (x - m)^2$, we obtain:

$$\mathrm{Var}(X) = \int_{\mathbb{R}} (x - m)^2\, dP_X(x).$$

If the distribution of $X$ has a density $f$, then:

$$\sigma^2 = \mathrm{Var}(X) = \int_{x\in\mathbb{R}} (x-m)^2 f(x)dx = \int_{x\in\mathbb{R}} x^2 f(x)dx - m^2.$$

**Example 2.6.2.**

- If $X \sim \mathrm{Exp}(\alpha)$, then $E(X) = \frac{1}{\alpha}$ and $\mathrm{Var}(X) = \frac{1}{\alpha^2}$.
- If $X \sim \mathcal{N}(m, \sigma^2)$, then $E(X) = m$ and $\mathrm{Var}(X) = \sigma^2$.

## 2.7 Dumb function method

The basic technique for determining the law of a random variable $X$ is the mute function method. We take a bounded Borel function $f : \mathbb{R} \to \mathbb{R}$ and evaluate $E(f(X))$. The transfer formula, $E(f(X)) = \int_{\mathbb{R}} f(x)\, dP_X(x)$, allows us to identify the law $P_X$. This method relies on the following proposition.

**Proposition 2.7.1.** Let $\mu_1, \mu_2$ be two probability measures on $\mathbb{R}$ such that for any continuous and bounded function $f : \mathbb{R} \to \mathbb{R}$, we have $\int_{\mathbb{R}} f d\mu_1 = \int_{\mathbb{R}} f d\mu_2$. Then $\mu_1 = \mu_2$.

**Example 2.7.2.** Let $X \sim \mathcal{U}([0,1])$ and $k \geq 1$. What is the distribution of $Y = X^k$?

**Solution.** Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous bounded function. Let's calculate $E(f(X^k))$.

$$\begin{aligned} E(f(X^k)) &= \int_\Omega f(X^k(\omega))dP(\omega) \\ &\overset{\text{transfert}}{=} \int_{\mathbb{R}} f(x^k)dP_X(x) && \text{with } P_X \text{ having density } \mathbb{1}_{[0,1]}(x) \\ &= \int_{\mathbb{R}} f(x^k)\mathbb{1}_{[0,1]}(x)dx = \int_0^1 f(x^k)dx \end{aligned}$$

Let's perform the change of variable $y = x^k$. Then $x = y^{1/k}$, and $dx = \frac{1}{k}y^{\frac{1}{k}-1}dy$. The integration limits remain from 0 to 1.

$$\begin{aligned} E(f(X^k)) &= \int_0^1 f(y) \cdot \frac{1}{k}y^{\frac{1}{k}-1}dy \\ &= \int_{\mathbb{R}} f(y)\left(\frac{1}{k}y^{\frac{1}{k}-1}\mathbb{1}_{[0,1]}(y)\right)dy \end{aligned}$$

We recognize the form $\int_{\mathbb{R}} f(y)g(y)dy$. By the transfer method, we deduce that the distribution of $X^k$ is a density distribution, and its density is the function $g(y) = \frac{1}{k}y^{\frac{1}{k}-1}\mathbb{1}_{[0,1]}(y)$. □

# CHAPTER 3

# INDEPENDENCE

## 3.1 Sigma algebra generated by a random variable

Let $(\Omega, \mathcal{F}, P)$ be a probability space.

**Definition 3.1.1** (Sub-$\sigma$-algebra)**.** A subtribe $\mathcal{G}$ of $\mathcal{F}$ is a *sigma*-algebra that is included in $\mathcal{F}$, that is, $\mathcal{G} \subset \mathcal{F}$. For any $A \in \mathcal{G}$, we have $A \in \mathcal{F}$.

**Definition 3.1.2** ($\sigma$-algebra of a Random Variable)**.** Let $X$ be a random variable (r.v.) from $(\Omega, \mathcal{F}, P)$ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. The sigma-algebra generated by the r.v. $X$, denoted by $\sigma(X)$, is the smallest sigma-algebra on $\Omega$ for which $X$ is measurable. It is defined as:

$$\sigma(X) = \sigma(\{X^{-1}(B) \mid B \in \mathcal{B}(\mathbb{R})\})$$

It should be noted that the collection of sets $\{X^{-1}(B) \mid B \in \mathcal{B}(\mathbb{R})\}$ is itself a sigma-algebra.

Since $X$ is a random variable, for all $B \in \mathcal{B}(\mathbb{R})$, we have $X^{-1}(B) \in \mathcal{F}$. Therefore, $\sigma(X)$ is a sub-*sigma*-algebra of $\mathcal{F}$.

## 3.2 Fundamental Definitions

### 3.2.1 Independence of events

**Definition 3.2.1.** Let $n \geq 2$. The $n$ events $A_1, \ldots, A_n$ in $\mathcal{F}$ are called independent if, for any subset $I \subseteq \{1, \ldots, n\}$, we have:

$$P\left(\bigcap_{i \in I} A_i\right) = \prod_{i \in I} P(A_i)$$

**Remark 3.2.2.** Note that "being independent" is a much stronger condition than "being pairwise independent."

### 3.2.2 Independence of random variables

**Definition 3.2.3.** The $n$ random variables $X_1, \ldots, X_n$ from $\Omega$ to $\mathbb{R}$ are said to be independent if, for all Borel sets $B_1, \ldots, B_n$ of $\mathbb{R}$, we have:

$$P(X_1 \in B_1, \ldots, X_n \in B_n) = P(X_1 \in B_1) \cdots P(X_n \in B_n)$$

where the notation $P(A, B)$ means $P(A \cap B)$.

### 3.2.3 Independence of sub-*sigma*-algebras

**Definition 3.2.4.** The $n$ subtribes $\mathcal{F}_1, \ldots, \mathcal{F}_n$ of $\mathcal{F}$ are called independent if, for any choice of events $A_1 \in \mathcal{F}_1, \ldots, A_n \in \mathcal{F}_n$, we have:

$$P(A_1 \cap \cdots \cap A_n) = P(A_1) \cdots P(A_n)$$

### 3.2.4 Links between the definitions

The three previous definitions are hierarchical.

- The events $A_1, \ldots, A_n$ are independent if and only if the indicator random variables $\mathbf{1}_{A_1}, \ldots, \mathbf{1}_{A_n}$ are independent.
- The random variables $X_1, \ldots, X_n$ are independent if and only if the *sigma*-algebras they generate, $\sigma(X_1), \ldots, \sigma(X_n)$, are independent.

The definition based on tribes is therefore the most general. It allows handling more complex situations, such as the case of random variables taking values in arbitrary spaces.

## 3.3 General random variables

**Definition 3.3.1.** Let $(E, \mathcal{E})$ be a measurable space. A random variable taking values in $E$ is a function $X : \Omega \to E$ that is measurable from $(\Omega, \mathcal{F})$ to $(E, \mathcal{E})$, that is:

$$\forall B \in \mathcal{E}, \quad X^{-1}(B) \in \mathcal{F}$$

**Definition 3.3.2** (Distribution of a Random Variable)**.** The law of $X$ is the probability measure $P_X$ on $(E, \mathcal{E})$ defined by:

$$\forall B \in \mathcal{E}, \quad P_X(B) = P(X \in B) = P(X^{-1}(B))$$

This formalism makes it possible to construct random vectors and even random functions.

**Example 3.3.3.** Let $X, Y$ be two random variables with values in $\mathbb{R}$. Let $Z = (X, Y)$. Then $Z$ is a random vector in $\mathbb{R}^2$. We say that the law of $Z$ admits a density if there exists a measurable function $f : \mathbb{R}^2 \to \mathbb{R}_+$ such that for all $B \in \mathcal{B}(\mathbb{R}^2)$:

$$P_Z(B) = P(Z \in B) = \iint_B f(x, y)\, dx\, dy$$

## 3.4 Product Laws

Let $(E_1, \mathcal{E}_1, \mu_1), \ldots, (E_n, \mathcal{E}_n, \mu_n)$ $n$ be probability spaces. A natural question arises: does there exist a probability space $(\Omega, \mathcal{F}, P)$ on which are defined $n$ random variables $X_1, \ldots, X_n$ such that:

- $X_1, \ldots, X_n$ are independent.

- For all $i \in \{1, \dots, n\}$, $X_i$ takes values in $E_i$ and has the distribution $\mu_i$.

The answer is affirmative and is provided by the construction of the product measure.

### 3.4.1 Construction of independent variables

Consider the Cartesian product space $E = E_1 \times \cdots \times E_n$. We equip it with the product sigma-algebra $\mathcal{E} = \mathcal{E}_1 \otimes \cdots \otimes \mathcal{E}_n$, which is the sigma-algebra generated by the "measurable rectangles" of the form $B_1 \times \cdots \times B_n$, where $B_i \in \mathcal{E}_i$ for each $i$.

**Theorem 3.4.1** (Product Law)**.** There exists a unique probability measure $\mu$ on the measurable space $(E, \mathcal{E})$ such that for all $B_1 \in \mathcal{E}_1, \dots, B_n \in \mathcal{E}_n$:

$$\mu(B_1 \times \cdots \times B_n) = \mu_1(B_1) \cdots \mu_n(B_n)$$

This measure is called the product measure of $\mu_1, \dots, \mu_n$ and is denoted by $\mu = \mu_1 \otimes \cdots \otimes \mu_n$.

To answer our initial question, we simply need to ask:

- $\Omega = E = E_1 \times \cdots \times E_n$
- $\mathcal{F} = \mathcal{E} = \mathcal{E}_1 \otimes \cdots \otimes \mathcal{E}_n$
- $P = \mu = \mu_1 \otimes \cdots \otimes \mu_n$

And to define the random variables $X_i$ as the projections onto the $i$-th component, for $i \in \{1, \dots, n\}$:

$$X_i : \Omega \to E_i, \quad \omega = (\omega_1, \dots, \omega_n) \mapsto \omega_i$$

**Proof.** Let us verify that these $X_i$ have the desired properties.

1. **Law of $X_i$:** Let $i \in \{1, \dots, n\}$ and $B_i \in \mathcal{E}_i$. The law of $X_i$ is given by:

$$\begin{aligned} P_{X_i}(B_i) = P(X_i \in B_i) &= \mu(\{\omega = (\omega_1, \dots, \omega_n) \in \Omega \mid \omega_i \in B_i\}) \\ &= \mu(E_1 \times \cdots \times E_{i-1} \times B_i \times E_{i+1} \times \cdots \times E_n) \\ &= \mu_1(E_1) \cdots \mu_{i-1}(E_{i-1}) \mu_i(B_i) \mu_{i+1}(E_{i+1}) \cdots \mu_n(E_n) \end{aligned}$$

   Since $\mu_j(E_j) = 1$ for all $j$, we obtain $P_{X_i}(B_i) = \mu_i(B_i)$. Thus, $X_i$ indeed has the law $\mu_i$.

2. **Independence of the $X_i$:** Let $B_1 \in \mathcal{E}_1, \dots, B_n \in \mathcal{E}_n$. By definition of independence:

$$\begin{aligned} P(X_1 \in B_1, \dots, X_n \in B_n) &= \mu(\{\omega = (\omega_1, \dots, \omega_n) \mid \forall i, \omega_i \in B_i\}) \\ &= \mu(B_1 \times \cdots \times B_n) \\ &= \mu_1(B_1) \cdots \mu_n(B_n) \\ &= P(X_1 \in B_1) \cdots P(X_n \in B_n) \end{aligned}$$

   The variables $X_1, \dots, X_n$ are therefore independent.

□

### 3.4.2 Application: i.i.d. variables

A particular important case is where $(E_i, \mathcal{E}_i) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ for every $i$, and where all laws are identical: $\mu_1 = \cdots = \mu_n = \nu$. The theorem then guarantees the existence of a probability space in which are defined $n$ random variables $X_1, \dots, X_n$ **independent and identically distributed** (i.i.d.), with common law $\nu$.

## 3.5 Grouping by blocks

Independence is maintained through grouping.

**Theorem 3.5.1.** Let $\mathcal{F}_1, \ldots, \mathcal{F}_n$ be independent sub-*sigma*-algebras. Let $I$ and $J$ be two disjoint subsets of $\{1, \ldots, n\}$. Then the *sigma*-algebras $\mathcal{G} = \sigma(\bigcup_{i \in I} \mathcal{F}_i)$ and $\mathcal{H} = \sigma(\bigcup_{j \in J} \mathcal{F}_j)$ are independent.

The proof of this theorem relies on the uniqueness extension lemma, also known as the monotone class lemma or Dynkin's $\pi - \lambda$ theorem.

**Lemma 3.5.2** (Uniqueness of the Extension)**.** Let $P_1$ and $P_2$ be two probability measures on a measurable space $(\Omega, \mathcal{F})$. Suppose there exists a collection $\mathcal{C}$ of events such that:

i) $\mathcal{C}$ generates $\mathcal{F}$, i.e., $\sigma(\mathcal{C}) = \mathcal{F}$.

ii) $\mathcal{C}$ is stable under finite intersection (it is a $\pi$-system), i.e., $\forall A, B \in \mathcal{C}, A \cap B \in \mathcal{C}$.

iii) $P_1$ and $P_2$ coincide on $\mathcal{C}$, i.e., $\forall C \in \mathcal{C}, P_1(C) = P_2(C)$.

Then $P_1 = P_2$ (they coincide on the entire $\sigma$-algebra $\mathcal{F}$).

**Proof** (Proof Idea of the Theorem)**.** On veut montrer que $\forall D \in \mathcal{G}, \forall E \in \mathcal{H}, P(D \cap E) = P(D)P(E)$.

1. On fixe d'abord un événement $D$ dans une classe simple génératrice de $\mathcal{G}$, par exemple $D = \bigcap_{i \in I} D_i$ avec $D_i \in \mathcal{F}_i$. On suppose $P(D) > 0$.

2. On définit deux mesures de probabilité sur $(\Omega, \mathcal{H})$ : $P_1(E) = P(E|D) = \frac{P(E \cap D)}{P(D)}$ et $P_2(E) = P(E)$.

3. On montre que $P_1$ et $P_2$ coïncident sur une collection $\mathcal{C}$ qui est un $\pi$-système engendrant $\mathcal{H}$. La collection des "rectangles" $E = \bigcap_{j \in J} E_j$ avec $E_j \in \mathcal{F}_j$ convient. En effet, pour un tel $E$ :

$$\begin{aligned} P(E \cap D) &= P\left(\left(\bigcap_{j \in J} E_j\right) \cap \left(\bigcap_{i \in I} D_i\right)\right) \\ &= \left(\prod_{j \in J} P(E_j)\right)\left(\prod_{i \in I} P(D_i)\right) \quad \text{(par indépendance des } \mathcal{F}_k) \\ &= P(E)P(D) \end{aligned}$$

   Donc $P_1(E) = P(E|D) = P(E) = P_2(E)$.

4. Par le lemme d'unicité, $P_1(E) = P_2(E)$ pour tout $E \in \mathcal{H}$, ce qui signifie $P(E \cap D) = P(E)P(D)$ pour tout $E \in \mathcal{H}$ et pour $D$ de la forme "rectangle".

5. On refait un raisonnement similaire en fixant $E \in \mathcal{H}$ et en faisant varier $D$ dans $\mathcal{G}$.

□

This theorem is generalized by recursion.

**Proposition 3.5.3.** Let $\mathcal{F}_1, \ldots, \mathcal{F}_n$ be independent *sigma*-algebras. Let $I_1, \ldots, I_k$ be subsets of $\{1, \ldots, n\}$ that are pairwise disjoint. Then the *sigma*-algebras $\mathcal{G}_j = \sigma(\bigcup_{i \in I_j} \mathcal{F}_i)$ for $j = 1, \ldots, k$ are independent.

A direct corollary for random variables is as follows:

**Theorem 3.5.4** (Corollary)**.** Let $X_1, \ldots, X_n$ be independent random variables. Let $I_1, \ldots, I_k$ be disjoint subsets of $\{1, \ldots, n\}$. The random vectors $V_j = (X_i)_{i \in I_j}$ for $j = 1, \ldots, k$ are independent.

## 3.6 Expectation and independence

Independence has a very important consequence for the calculation of expectations.

**Proposition 3.6.1.** Let $X$ and $Y$ be two real random variables. The following two conditions are equivalent:

i) $X$ and $Y$ are independent.

ii) For all Borel functions $f, g : \mathbb{R} \to \mathbb{R}$ such that $f(X)$, $g(Y)$, and $f(X)g(Y)$ are integrable, we have:

$$E[f(X)g(Y)] = E[f(X)]E[g(Y)]$$

**Proof.** **(ii) $\implies$ (i) :** We just need to choose indicator functions. Let $A, B \in \mathcal{B}(\mathbb{R})$. We set $f = \mathbf{1}_A$ and $g = \mathbf{1}_B$. These functions are Borel measurable and bounded, so the random variables $\mathbf{1}_A(X)$, $\mathbf{1}_B(Y)$ and their product are integrable. The equality becomes:

$$E[\mathbf{1}_A(X)\mathbf{1}_B(Y)] = E[\mathbf{1}_A(X)]E[\mathbf{1}_B(Y)]$$

Which translates to:

$$P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$$

Since this is true for all Borel sets $A$ and $B$, $X$ and $Y$ are independent. **(i) $\implies$ (ii) :** The proof proceeds by successive steps (monotone class theorem).

1. **Indicator functions :** If $f = \mathbf{1}_A$ and $g = \mathbf{1}_B$, the equality is the definition of independence, as seen above. It is therefore true.

2. **Positive simple functions :** By linearity of expectation, the equality extends to positive simple functions (finite linear combinations with positive coefficients of indicator functions).

3. **Positive measurable functions :** Any positive measurable function can be approximated by a non-decreasing sequence of positive simple functions. By the monotone convergence theorem, the equality holds in the limit.

4. **General integrable functions :** We decompose $f = f^+ - f^-$ and $g = g^+ - g^-$, where $f^+, f^-, g^+, g^-$ are positive functions. We apply the previous result to each pair and use the linearity of expectation.

$$\begin{aligned}
E[f(X)g(Y)] &= E[(f^+(X) - f^-(X))(g^+(Y) - g^-(Y))] \\
&= E[f^+(X)g^+(Y) - f^+(X)g^-(Y) - f^-(X)g^+(Y) + f^-(X)g^-(Y)] \\
&= E[f^+(X)g^+(Y)] - E[f^+(X)g^-(Y)] - E[f^-(X)g^+(Y)] + E[f^-(X)g^-(Y)] \\
&= E[f^+(X)]E[g^+(Y)] - E[f^+(X)]E[g^-(Y)] - E[f^-(X)]E[g^+(Y)] + E[f^-(X)]E[g^-(Y)] \\
&= (E[f^+(X)] - E[f^-(X)])(E[g^+(Y)] - E[g^-(Y)]) \\
&= E[f(X)]E[g(Y)]
\end{aligned}$$

□

An immediate and fundamental corollary:

**Theorem 3.6.2** (Corollary)**.** Let $X$ and $Y$ be two independent random variables such that $E[|X|] < \infty$ and $E[|Y|] < \infty$. Then $XY$ is integrable ($E[|XY|] < \infty$) and we have:

$$E[XY] = E[X]E[Y]$$

## 3.7 Covariance

**Definition 3.7.1.** Let $X$ and $Y$ be two random variables on $(\Omega, \mathcal{F}, P)$ that are integrable, and whose product $XY$ is also integrable. The covariance of $X$ and $Y$ is defined by:

$$\operatorname{cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

By developing the expression, we obtain a more practical formula:

$$\begin{aligned}\operatorname{cov}(X, Y) &= E[XY - XE[Y] - YE[X] + E[X]E[Y]] \\ &= E[XY] - E[X]E[Y] - E[Y]E[X] + E[X]E[Y] \\ &= E[XY] - E[X]E[Y]\end{aligned}$$

Covariance is a measure of the linear dependence between two variables.

**Proposition 3.7.2.** If $X$ and $Y$ are independent and integrable, then their covariance is zero:

$$\operatorname{cov}(X, Y) = 0$$

**Proof.** If $X$ and $Y$ are independent and integrable, by the previous corollary, $E[XY] = E[X]E[Y]$. Consequently, $\operatorname{cov}(X, Y) = E[XY] - E[X]E[Y] = 0$. □

**Remark 3.7.3.** The converse is false! Two variables may have zero covariance (they are said to be uncorrelated) without being independent. Covariance only captures linear dependencies.

# CHAPTER 4

# INFINITE SEQUENCES OF RANDOM VARIABLES

Let $(\Omega, \mathcal{F}, P)$ be a probability space.

## 4.1 Convergence modes

Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of random variables (r.v.) defined on $(\Omega, \mathcal{F}, P)$. Let also $X$ be another r.v. defined on $(\Omega, \mathcal{F}, P)$.

**Definition 4.1.1** (Almost Sure Convergence)**.** The sequence $(X_n)_{n\in\mathbb{N}}$ converges almost surely to $X$, which we denote by $X_n \xrightarrow{p.s.} X$, if

$$P(\{\omega \in \Omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\}) = 1.$$

**Definition 4.1.2** (Convergence in Mean of Order $r$ ($L^r$))**.** Let $r \geq 1$. The sequence $(X_n)_{n\in\mathbb{N}}$ converges to $X$ in $L^r(\Omega, P)$, denoted by $X_n \xrightarrow{L^r} X$, if

$$\lim_{n\to\infty} \mathbb{E}[|X_n - X|^r] = 0.$$

**Definition 4.1.3** (Convergence in Probability)**.** The sequence $(X_n)_{n\in\mathbb{N}}$ converges to $X$ in probability, denoted as $X_n \xrightarrow{P} X$, if

$$\forall \varepsilon > 0, \quad \lim_{n\to\infty} P(|X_n - X| > \varepsilon) = 0.$$

## 4.2 Links between convergences

The most difficult convergence to establish is almost sure convergence.

**Proposition 4.2.1.** If $X_n \xrightarrow{p.s.} X$, then $X_n \xrightarrow{P} X$.

**Proof.** Assume $X_n \xrightarrow{p.s.} X$. By definition, this means that $P(\{\omega \in \Omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\}) = 1$.

The definition of the limit of a sequence is:

$$\lim_{n\to\infty} X_n(\omega) = X(\omega) \iff \forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall n > N, |X_n(\omega) - X(\omega)| < \varepsilon.$$

For measurability reasons, we restrict $\varepsilon$ to positive rationals, which is sufficient.

$$\begin{aligned} P(\lim_{n\to\infty} X_n = X) = 1 &\iff P\left(\forall \varepsilon \in \mathbb{Q}_+^*, \exists N \in \mathbb{N}, \forall n > N, |X_n - X| < \varepsilon\right) = 1 \\ &\iff P\left(\bigcap_{\varepsilon\in\mathbb{Q}_+^*} \bigcup_{N\in\mathbb{N}} \bigcap_{n>N} \{|X_n - X| < \varepsilon\}\right) = 1 \end{aligned}$$

Since this is a countable intersection of events with probability 1, each of these events must have a probability of 1.

$$\implies \forall \varepsilon > 0, \quad P\left(\bigcup_{N\in\mathbb{N}} \bigcap_{n>N} \{|X_n - X| < \varepsilon\}\right) = 1$$

Taking the complement, the probability becomes 0:

$$\forall \varepsilon > 0, \quad P\left(\left(\bigcup_{N\in\mathbb{N}} \bigcap_{n>N} \{|X_n - X| < \varepsilon\}\right)^c\right) = P\left(\bigcap_{N\in\mathbb{N}} \bigcup_{n>N} \{|X_n - X| \geq \varepsilon\}\right) = 0$$

Let $A_N = \bigcup_{n>N}\{|X_n - X| \geq \varepsilon\}$. The sequence of events $(A_N)_{N\in\mathbb{N}}$ is decreasing. By continuity of the probability measure for decreasing sequences, we have:

$$P\left(\bigcap_{N\in\mathbb{N}} A_N\right) = \lim_{N\to\infty} P(A_N) = \lim_{N\to\infty} P\left(\bigcup_{n>N} \{|X_n - X| \geq \varepsilon\}\right) = 0.$$

However, for all $N$, we have the inclusion $\{|X_N - X| \geq \varepsilon\} \subset \bigcup_{n>N}\{|X_n - X| \geq \varepsilon\}$. Therefore, $P(|X_N - X| \geq \varepsilon) \leq P(\bigcup_{n>N}\{|X_n - X| \geq \varepsilon\})$. Taking the limit as $N \to \infty$, we obtain:

$$\forall \varepsilon > 0, \quad \lim_{N\to\infty} P(|X_N - X| \geq \varepsilon) = 0.$$

This is the definition of convergence in probability, $X_n \xrightarrow{P} X$. □

**Proposition 4.2.2.** Let $r \geq 1$. If $X_n \xrightarrow{L^r} X$, then $X_n \xrightarrow{P} X$.

**Proof.** Let $\varepsilon > 0$. We use Markov's inequality. Let's write:

$$\begin{aligned} \mathbb{E}[|X_n - X|^r] &= \int_\Omega |X_n - X|^r dP \\ &= \int_{\{|X_n - X| \geq \varepsilon\}} |X_n - X|^r dP + \int_{\{|X_n - X| < \varepsilon\}} |X_n - X|^r dP \\ &\geq \int_{\{|X_n - X| \geq \varepsilon\}} |X_n - X|^r dP \\ &\geq \int_{\{|X_n - X| \geq \varepsilon\}} \varepsilon^r dP = \varepsilon^r P(|X_n - X| \geq \varepsilon). \end{aligned}$$

From this, we obtain:

$$P(|X_n - X| > \varepsilon) \leq \frac{1}{\varepsilon^r}\mathbb{E}[|X_n - X|^r].$$

Since $X_n \xrightarrow{L^r} X$, we have $\lim_{n\to\infty} \mathbb{E}[|X_n - X|^r] = 0$. Therefore,

$$\lim_{n\to\infty} P(|X_n - X| > \varepsilon) = 0,$$

which proves that $X_n \xrightarrow{P} X$. □

**Proposition 4.2.3** (Compatibility with Addition)**.** If $X_n \xrightarrow{*} X$ and $Y_n \xrightarrow{*} Y$, then $X_n + Y_n \xrightarrow{*} X + Y$, where $*$ can be almost sure convergence, $L^r$ (with the same $r \geq 1$), or convergence in probability.

**Proof** (Hint for Convergence in Probability)**.** We use the triangle inequality and the sub-additivity of probability. For all $\varepsilon > 0$:

$$\{|(X_n + Y_n) - (X + Y)| > \varepsilon\} \subset \{|X_n - X| + |Y_n - Y| > \varepsilon\} \subset \{|X_n - X| > \varepsilon/2\} \cup \{|Y_n - Y| > \varepsilon/2\}.$$

Hence,

$$P(|(X_n + Y_n) - (X + Y)| > \varepsilon) \leq P(|X_n - X| > \varepsilon/2) + P(|Y_n - Y| > \varepsilon/2).$$

Each of the terms on the right tends to 0 as $n \to \infty$. □

We can also translate classical measure theory theorems into the language of random variables.

**Example 4.2.4** (Dominated Convergence Theorem)**.** Let $(X_n)$ be a sequence of random variables such that:

1. There exists a random variable $X$ such that $X_n \xrightarrow{p.s.} X$.
2. There exists a random variable $Y \geq 0$ that is integrable ($\mathbb{E}[Y] < \infty$) such that $|X_n| \leq Y$ almost surely for all $n$.

Then $X_n \xrightarrow{L^1} X$, and in particular, $\lim_{n\to\infty} \mathbb{E}[X_n] = \mathbb{E}[X]$.

## 4.3 Examples

We consider the probability space $(\Omega, \mathcal{F}, P) = ([0,1], \mathcal{B}([0,1]), \lambda_{|[0,1]})$, where $\lambda$ is the Lebesgue measure.

**Example 4.3.1.** Consider the sequence of functions defined by $X_{2k}(\omega) = \mathbf{1}_{[0,1/2]}(\omega)$ and $X_{2k+1}(\omega) = \mathbf{1}_{[1/2,1]}(\omega)$ for $k \in \mathbb{N}$. For any $\omega \in [0,1]$, the sequence $(X_n(\omega))$ oscillates between 0 and 1 (except at $\omega = 1/2$) and thus does not converge. This sequence does not converge in any sense (a.s., $L^r$, or P).

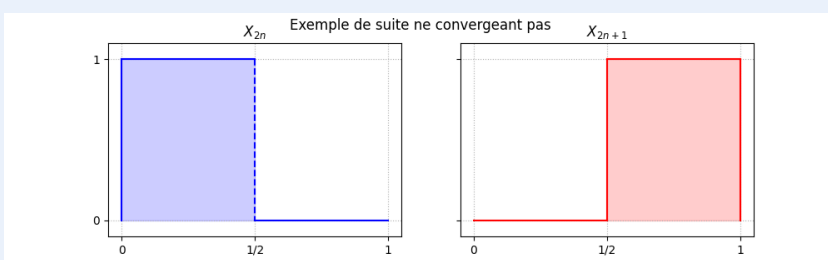



Figure 4.1: Sequence $(X_n)$ that has no limit.

**Example 4.3.2** (The "Typewriter" Sequence)**.** Consider the sequence of random variables $(X_n)$ which are indicators of intervals that sweep $[0,1]$ with a width that tends to 0. For example: $X_1 = \mathbf{1}_{[0,1]}$, $X_2 = \mathbf{1}_{[0,1/2]}$, $X_3 = \mathbf{1}_{[1/2,1]}$, $X_4 = \mathbf{1}_{[0,1/3]}$, etc. This sequence converges in probability to 0, as the measure (length) of the interval where $X_n = 1$ tends to 0.

$$P(|X_n - 0| > \varepsilon) = P(X_n = 1) \to 0 \quad \text{pour } 0 < \varepsilon < 1.$$

However, for every $\omega \in [0,1]$, the sequence $X_n(\omega)$ takes the value 1 infinitely many times. Therefore, $\limsup_{n\to\infty} X_n(\omega) = 1$, while $\liminf_{n\to\infty} X_n(\omega) = 0$. The pointwise limit does not exist, so the sequence does not converge almost surely. We thus have $X_n \xrightarrow{P} 0$ but $X_n \not\xrightarrow{p.s.} 0$.

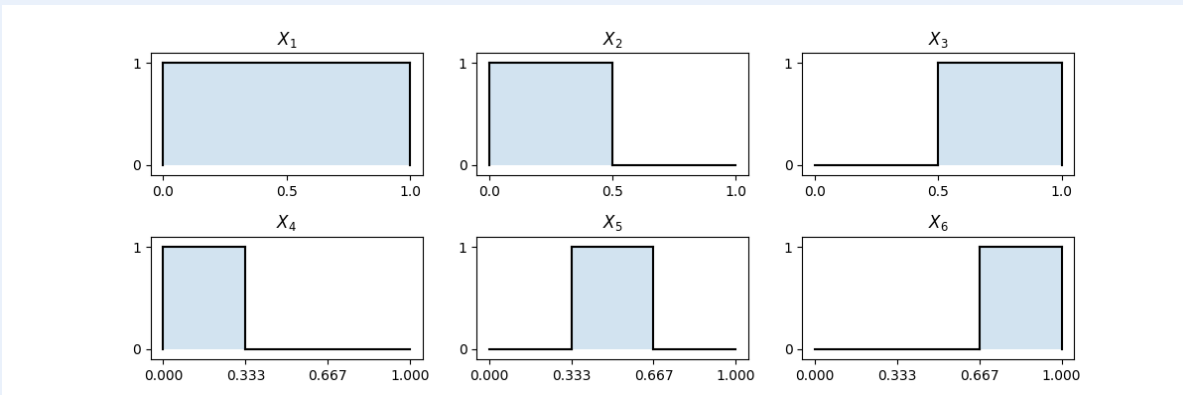



Figure 4.2: The first terms of the "typewriter" sequence.

## 4.4 Independence for an infinite sequence

**Definition 4.4.1.** An infinite sequence of *sigma*-fields $(\mathcal{F}_n)_{n\in\mathbb{N}}$ is said to be independent (resp. of events, of random variables) if every finite family extracted from the sequence is independent.

**Remark 4.4.2.** There is an analogy with the concept of freedom in linear algebra.

**Theorem 4.4.3** (Grouping by Infinite Blocks)**.** Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of independent random variables. Let $(I_k)_{k\in\mathbb{N}}$ be a partition of $\mathbb{N}$ (pairwise disjoint sets whose union is $\mathbb{N}$). Then the *sigma*-algebras $\mathcal{J}_k = \sigma(X_n, n \in I_k)$ for $k \in \mathbb{N}$ are independent.

## 4.5 The analytical model for the coin toss game

We want to construct a probability space $(\Omega, \mathcal{F}, P)$ on which a sequence of random variables $(X_n)_{n\in\mathbb{N}^*}$ with a symmetric Bernoulli distribution on $\{-1,1\}$ (sometimes called the Rademacher distribution) is defined, and which are all independent. We take $(\Omega, \mathcal{F}, P) = ([0,1], \mathcal{B}([0,1]), \lambda)$, the Lebesgue measure on $[0,1]$. To a real number $\omega \in [0,1[$, we associate its unique dyadic (binary) expansion, the one that does not end with an infinite sequence of 1s:

$$\omega = \sum_{n=1}^{\infty} \frac{\omega_n}{2^n}, \quad \text{where } \omega_n \in \{0,1\}.$$

We then define, for $n \geq 1$, the sequence of random variables $(X_n)$ as follows:

$$\forall \omega \in [0,1[, \quad X_n(\omega) = 2\omega_n - 1 \in \{-1,1\}.$$

These functions are called the Rademacher functions. It can be shown that this sequence $(X_n)$ answers the question. For example, $X_1(\omega) = 1$ if $\omega_1 = 1$ (i.e., $\omega \in [1/2, 1[$) and $X_1(\omega) = -1$ if $\omega_1 = 0$ (i.e., $\omega \in [0, 1/2[$). In general, the set $X_n^{-1}(\{1\})$ is the union of $2^{n-1}$ dyadic intervals of length $1/2^n$. The total measure of this set is thus $2^{n-1} \times \frac{1}{2^n} = \frac{1}{2}$Hence $P(X_n = 1) = \lambda(\{\omega : X_n(\omega) = 1\}) = \frac{1}{2}$, and similarly $P(X_n = -1) = \frac{1}{2}$.

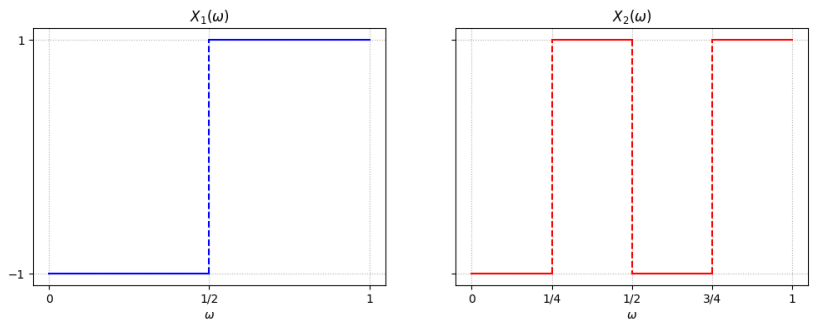



Figure 4.3: The first two Rademacher functions.

To show that $(X_n)_{n\geq 1}$ are independent, it suffices to show that for all $\varepsilon_1, \ldots, \varepsilon_n \in \{-1, 1\}$, we have $P(X_1 = \varepsilon_1, \ldots, X_n = \varepsilon_n) = \prod_{i=1}^n P(X_i = \varepsilon_i)$.

$$\begin{aligned} P(X_1 = \varepsilon_1, \ldots, X_n = \varepsilon_n) &= \lambda(\{\omega \in [0, 1[ : \forall i \in \{1..n\}, X_i(\omega) = \varepsilon_i\}) \\ &= \lambda(\{\omega \in [0, 1[ : \forall i \in \{1..n\}, \omega_i = \frac{1 + \varepsilon_i}{2}\}) \end{aligned}$$

The set of all $\omega$ whose first $n$ binary digits are fixed forms a dyadic interval of length $1/2^n$. For example, if $\varepsilon_1 = 1, \varepsilon_2 = -1$, we look for the $\omega$ with $\omega_1 = 1, \omega_2 = 0$, that is, $\omega \in [0.10_2, 0.11_2) = [1/2, 3/4[$. The length is $1/4 = 1/2^2$. Therefore, $P(X_1 = \varepsilon_1, \ldots, X_n = \varepsilon_n) = \frac{1}{2^n}$. Since $P(X_i = \varepsilon_i) = 1/2$ for all $i$, we indeed have $\prod_{i=1}^n P(X_i = \varepsilon_i) = (\frac{1}{2})^n$. The variables are therefore independent. We have an infinite sequence of independent "coins".

## 4.6 Random walk on the integers

We model a symmetric random walk on $\mathbb{Z}$.

- The walk starts at 0.
- At each step, a fair coin is flipped.
- Heads: take a step of +1 (to the right). Tails: take a step of -1 (to the left).
- The operation is repeated from the new position.

Let us denote $S_n$ as the position of the walker after $n$ steps. Formally:

$$S_0 = 0, \quad S_n = X_1 + X_2 + \cdots + X_n,$$

where the $(X_i)$ are a sequence of i.i.d. random variables with $P(X_i = 1) = P(X_i = -1) = 1/2$.

$$\text{"Lighter"} \quad \longleftrightarrow \quad 0 \quad \longleftrightarrow \quad \text{"Streetlight"}$$

A natural question is: what is the probability that the walker will one day pass by its starting point (the "original lamppost") again? We therefore seek to calculate:

$$P(\exists n \in \mathbb{N}^*, S_n = 0) = P\left(\bigcup_{n=1}^{\infty} \{S_n = 0\}\right).$$

For $S_n = 0$ to occur, there must be as many +1 steps as -1 steps. The total number of steps $n$ must therefore be even. Let us set $n = 2k$. First, we calculate the probability of returning to 0 after $2k$ steps. Let $\alpha$ be the number of +1 steps and $\beta$ the number of -1 steps. We must have $\alpha + \beta = 2k$ and $\alpha - \beta = 0$, which implies $\alpha = \beta = k$. The number of paths of length$2k$ returning to the origin is the number of ways to choose the $k$ steps +1 among the $2k$ steps, which is $\binom{2k}{k}$. The total number of possible paths of length $2k$ is $2^{2k}$. Thus,

$$P(S_{2k} = 0) = \frac{\binom{2k}{k}}{2^{2k}}.$$

Using Stirling's approximation, $m! \sim \sqrt{2\pi m}(m/e)^m$, we find:

$$P(S_{2k} = 0) \approx \frac{1}{\sqrt{\pi k}} \quad \text{for } k \to \infty.$$

The probability of returning to the origin at a given time tends towards 0.

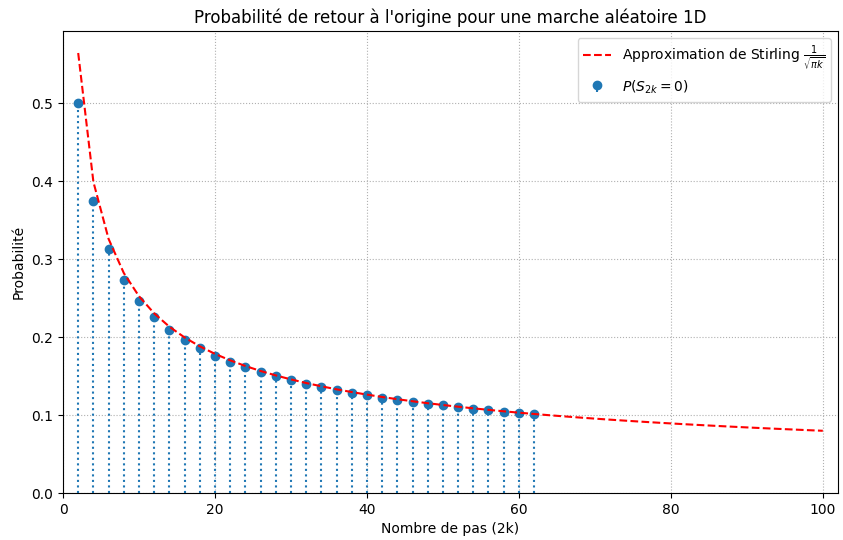



Figure 4.4: Evolution of $P(S_{2k} = 0)$ as a function of the number of steps.

To calculate the probability of returning *one day*, we decompose the event into a union of disjoint events: the first return to the origin. Let $f_{2k} = P(S_2 \neq 0, \ldots, S_{2k-2} \neq 0, S_{2k} = 0)$. The probability of returning one day is $\sum_{k=1}^{\infty} f_{2k}$.

**Lemma 4.6.1.** The probability of first return at time $2k$ is related to the return probability by the (non-trivial) relation:

$$f_{2k} = \frac{1}{2k-1} P(S_{2k} = 0) \quad (\text{for } k \geq 1).$$

The probability of returning one day is therefore $\sum_{k=1}^{\infty} f_{2k}$. It turns out that this sum equals 1.

$$P(\exists n \in \mathbb{N}^*, S_n = 0) = \sum_{k=1}^{\infty} f_{2k} = 1.$$

The simple symmetric random walk on $\mathbb{Z}$ is therefore *recurrent*: it is certain to return to its starting point.

# CHAPTER 5

# TRANSIENCE AND RECURRENCE OF WALKS

Let $d \geq 1$ and $(X_n)_{n\geq 1}$ be a sequence of random variables defined on $(\Omega, \mathcal{F}, P)$ with values in $\mathbb{Z}^d$. The associated random walk is the sequence $(S_n)_{n\geq 0}$ defined by $S_0 = 0, \quad \forall n \geq 1 \quad S_n = X_1 + \cdots + X_n$.

The random walk $(S_n)_{n\geq 0}$ is said to be recurrent if it returns to 0 infinitely often with probability 1, and transient otherwise.

## 5.1 First recurrence criterion

**Proposition 5.1.1.** If the walk returns to 0 with probability 1, then it passes through 0 infinitely often with probability 1.

**Proof.** Let $R$ be the time of the last visit to 0, i.e., $R = \sup\{n > 0; S_n = 0\}$. Assume that $P(\exists n \geq 1, S_n = 0) = 1$ and consider, for $k \geq 1$

$$\begin{aligned} P(R = k) &= P(S_k = 0, \forall n > k, S_n - S_k \neq 0) \\ &= P(X_1 + \cdots + X_k = 0, \forall n > k, X_{k+1} + \cdots + X_n \neq 0) \end{aligned}$$

The event $\{\forall n > k, X_{k+1} + \cdots + X_n \neq 0\}$ is in $\sigma(X_i, i \geq k+1)$. By the blocking theorem, this event is independent of $\{X_1 + \cdots + X_k = 0\}$. Hence, $P(R = k) = P(S_k = 0)P(\forall n > k, S_n - S_k \neq 0)$. Moreover, the sequence $(X_{k+1}, X_{k+2}, \dots)$ has the same distribution as $(X_1, X_2, \dots)$. Thus, $P(\forall n > k, S_n - S_k \neq 0) = P(\forall n \geq 1, S_n \neq 0)$. Given that $P(S_k = 0) \leq 1$, we have:

$$\begin{aligned} P(R = k) &\leq P(\forall n \geq 1, S_n \neq 0) \\ &\leq 1 - P(\exists n \geq 1, S_n = 0) = 1 - 1 = 0 \end{aligned}$$

Thus, $P(R = k) = 0$ for all $k \in \mathbb{N}^*$. Now, the probability that the walk visits 0 a finite number of times is:

$$P(R < +\infty) = P(\bigcup_{k\in\mathbb{N}^*} \{R = k\}) \leq \sum_{k\in\mathbb{N}^*} P(R = k) = 0$$

This means that the probability that the walk returns only a finite number of times is zero. Therefore, it returns an infinite number of times with probability 1. □

**Theorem 5.1.2** (Equivalence)**.** There is equivalence between:

1. $P(\exists n \geq 1, S_n = 0) = 1$ (the walk passes back through 0).

2. $P(S_n$ visits 0 infinitely often$) = 1$.

**Remark 5.1.3.** Often, it is easier to show that an event occurs infinitely often with probability 1 rather than showing that it occurs at least once with probability 1.

## 5.2 The Borel-Cantelli lemma

Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let $(A_n)_{n\in\mathbb{N}}$ be a sequence of events in $\mathcal{F}$. We define the lower limit and the upper limit of this sequence of events.

**Definition 5.2.1.**
- $\liminf A_n = \bigcup_{n\geq 0}\bigcap_{m\geq n} A_m = \{\omega \in \Omega : \omega$ belongs to all $A_n$ except for a finite number$\}$.
- $\limsup A_n = \bigcap_{n\geq 0}\bigcup_{m\geq n} A_m = \{\omega \in \Omega : \omega$ belongs to infinitely many $A_n\}$.

Moreover, the event $\limsup A_n$ is often denoted $\{A_n$ occurs infinitely often$\}$ or, in abbreviated form, $\{A_n$ i.s.$\}$ (infinitely often).

**Example 5.2.2.**
- Let $(X_n)$ be a sequence of real random variables and $B \in \mathcal{B}(\mathbb{R})$. Then $\limsup\{X_n \in B\} = \{X_n \in B$ a.s.$\}$.
- For the random walk, $\limsup\{S_n = 0\} = \{S_n = 0$ a.s.$\}$.

**Lemma 5.2.3** (Borel-Cantelli Lemma)**.** Let $(A_n)_{n\in\mathbb{N}}$ be a sequence of events.

- **Part I:** If the series $\sum_n P(A_n)$ converges, then $P(A_n$ i.s.$) = 0$.
- **Part II:** If the series $\sum_n P(A_n)$ diverges and the events $(A_n)$ are independent, then $P(A_n$ i.s.$) = 1$.

**Proof. I)** Assume that the series $\sum P(A_n)$ converges. Then

$$P(A_n \text{ occurs infinitely often}) = P(\bigcap_{n\geq 0}\bigcup_{m\geq n} A_m)$$

The sequence of events $B_n = \bigcup_{m\geq n} A_m$ is decreasing. By the monotone convergence theorem:

$$P(\bigcap_{n\geq 0} B_n) = \lim_{n\to\infty} P(B_n) = \lim_{n\to\infty} P(\bigcup_{m\geq n} A_m)$$

By $\sigma$-additivity:

$$P(\bigcup_{m\geq n} A_m) \leq \sum_{m\geq n} P(A_m)$$

As the series $\sum P(A_n)$ converges, the remainder of the series tends to 0:

$$\lim_{n\to\infty}\sum_{m\geq n} P(A_m) = 0$$

Thus, $P(A_n$ i.o.$) = 0$. **II)** Assume that $\sum P(A_n) = +\infty$ and that the $(A_n)$ are independent. We want

to show that $P(A_n \text{ i.o.}) = P(\bigcap_n \bigcup_{m\geq n} A_m) = 1$. Consider the complement:

$$P(\{A_n \text{ i.o.}\}^c) = P(\bigcup_n \bigcap_{m\geq n} A_m^c)$$

Let $n$ be fixed, consider $P(\bigcap_{m\geq n} A_m^c)$. For any $N > n$:

$$P(\bigcap_{m=n}^{N} A_m^c) = \prod_{m=n}^{N} P(A_m^c) \quad \text{by independence}$$

$$= \prod_{m=n}^{N} (1 - P(A_m))$$

Using the fact that $1 - x \leq e^{-x}$, we have

$$\prod_{m=n}^{N} (1 - P(A_m)) \leq \prod_{m=n}^{N} e^{-P(A_m)} = \exp\left(-\sum_{m=n}^{N} P(A_m)\right)$$

Since the series $\sum P(A_n)$ diverges, $\sum_{m=n}^{N} P(A_m) \to \infty$ when $N \to \infty$. Thus, $\exp(-\sum_{m=n}^{N} P(A_m)) \to 0$ when $N \to \infty$. Therefore, for all $n$, $P(\bigcap_{m\geq n} A_m^c) = 0$. By $\sigma$-additivity,

$$P(\{A_n \text{ i.o.}\}^c) = P(\bigcup_n \bigcap_{m\geq n} A_m^c) \leq \sum_n P(\bigcap_{m\geq n} A_m^c) = \sum_n 0 = 0.$$

Hence $P(A_n \text{ i.o.}) = 1$. □

### 5.2.1 Application: The monkey and Shakespeare

Let's put a monkey to type on a typewriter. What is the probability that it types the complete works of Shakespeare? Let $(X_n)_{n\geq 1}$ be an i.i.d sequence of Bernoulli(p), $0 < p < 1$. $\forall n \geq 1$, $P(X_n = 1) = p$, $P(X_n = 0) = 1 - p$. Let $k \geq 1$ and $(\epsilon_1, \ldots, \epsilon_k)$ be a fixed sequence of 0s and 1s. Consider the event $A_n = \{X_n = \epsilon_1, X_{n+1} = \epsilon_2, \ldots, X_{n+k-1} = \epsilon_k\}$. The probability of this event is $P(A_n) = p^{\sum \epsilon_i}(1-p)^{k-\sum \epsilon_i}$, which does not depend on $n$. Thus, the series $\sum P(A_n) = +\infty$. However, the $(A_n)$are not independent, we cannot apply part II of Borel-Cantelli directly. Consider the events $B_n = A_{nk}$, for $n \geq 1$. By the block grouping theorem, the $(B_n)$ are independent. And $\sum P(B_n) = +\infty$. By BC II, $P(B_n \text{ a.s.}) = 1$, which implies that the event occurs infinitely often.

## 5.3 A second recurrence criterion

Let us return to our random walk $S_0 = 0$, $S_n = X_1 + \cdots + X_n$. Let $T$ be the time of the first return to 0: $T = \min\{n \geq 1, S_n = 0\}$, with the convention $\inf \emptyset = +\infty$. For $n \geq 1$, let $u_n = P(S_n = 0)$ and $f_n = P(T = n)$. We set $u_0 = 1$ and $f_0 = 0$. For $n \geq 1$, we can decompose the event $\{S_n = 0\}$ according to the value of the first return time $T$:

$$u_n = P(S_n = 0) = P(\{S_n = 0\} \cap \bigcup_{k=1}^{n} \{T = k\}) = \sum_{k=1}^{n} P(S_n = 0, T = k)$$

$$= \sum_{k=1}^{n} P(S_1 \neq 0, \ldots, S_{k-1} \neq 0, S_k = 0, S_n = 0)$$

By the independence of increments, the event $\{S_n - S_k = 0\}$ is independent of $\{T = k\}$.

$$\begin{aligned} u_n &= \sum_{k=1}^{n} P(S_1 \neq 0, \ldots, S_{k-1} \neq 0, S_k = 0) P(S_n - S_k = 0) \\ &= \sum_{k=1}^{n} P(T = k) P(S_{n-k} = 0) \\ u_n &= \sum_{k=1}^{n} f_k u_{n-k} \quad \forall n \geq 1 \end{aligned}$$

To the sequences $(u_n)$ and $(f_n)$, we associate their generating series, defined for $s \in [0, 1)$:

$$\begin{aligned} U(s) &= \sum_{n \geq 0} u_n s^n \\ F(s) &= \sum_{n \geq 0} f_n s^n = E(s^T) \end{aligned}$$

The recurrence relation $u_n = \sum_{k=1}^{n} f_k u_{n-k}$ is a convolution. For $n \geq 1$:

$$u_n s^n = \sum_{k=1}^{n} f_k u_{n-k} s^n = \sum_{k=1}^{n} (f_k s^k)(u_{n-k} s^{n-k})$$

By summing over $n \geq 1$:

$$\begin{aligned} \sum_{n \geq 1} u_n s^n &= \sum_{n \geq 1} \sum_{k=1}^{n} (f_k s^k)(u_{n-k} s^{n-k}) \\ U(s) - u_0 &= \left( \sum_{k \geq 1} f_k s^k \right) \left( \sum_{j \geq 0} u_j s^j \right) \\ U(s) - 1 &= F(s) U(s) \end{aligned}$$

Therefore, we obtain the relation $U(s)(1 - F(s)) = 1$ for $s \in [0, 1)$. When letting $s \to 1^-$, since the series has positive terms, we have by the monotone convergence theorem:

$$\begin{aligned} \lim_{s \to 1^-} U(s) &= \sum_{n \geq 0} u_n = \sum_{n \geq 0} P(S_n = 0) \\ \lim_{s \to 1^-} F(s) &= \sum_{n \geq 0} f_n = \sum_{n \geq 0} P(T = n) = P(T < +\infty) = P(\exists n \geq 1, S_n = 0) \end{aligned}$$

Thus, we obtain the relation:

$$\left( \sum_{n \geq 0} P(S_n = 0) \right) (1 - P(\exists n \geq 1, S_n = 0)) = 1$$

This equality implies that both terms are either finite or infinite at the same time. In particular, $\sum P(S_n = 0) = +\infty$ if and only if $1 - P(\exists n \geq 1, S_n = 0) = 0$, that is, $P(\exists n \geq 1, S_n = 0) = 1$. This leads us to the following equivalence theorem:

**Theorem 5.3.1** (Recurrence Criteria)**.** The following statements are equivalent:

1. $P(\exists n \geq 1, S_n = 0) = 1$ (the walk is recurrent in the sense of the first return).
2. $P(S_n = 0 \text{ i.s.}) = 1$ (the walk is recurrent).

3. $\sum_{n=0}^{\infty} P(S_n = 0) = +\infty$.

## 5.4 Application to walks on $\mathbb{Z}^d$

### 5.4.1 The walk on $\mathbb{Z}$

Let $0 < p < 1$ and let $(X_n)$ be i.i.d. with law $\mathrm{Ber}(p)$ on $\{-1, 1\}$, i.e., $P(X_n = 1) = p$ and $P(X_n = -1) = 1-p$. For $S_{2n} = 0$ to occur, there must be as many steps of $+1$ as steps of $-1$, i.e., $n$ of each. $P(S_{2k+1} = 0) = 0$.

$$P(S_{2n} = 0) = \binom{2n}{n} p^n (1-p)^n$$

Using Stirling's approximation $k! \sim \sqrt{2\pi k}(\frac{k}{e})^k$, we obtain:

$$\binom{2n}{n} \sim \frac{4^n}{\sqrt{\pi n}}$$

Hence,

$$P(S_{2n} = 0) \sim \frac{(4p(1-p))^n}{\sqrt{\pi n}}$$

The series $\sum_n P(S_{2n} = 0)$ behaves like a geometric series.

- If $p = 1/2$, $4p(1-p) = 1$ and $P(S_{2n} = 0) \sim \frac{1}{\sqrt{\pi n}}$. The series $\sum \frac{1}{\sqrt{n}}$ diverges, so the symmetric walk on $\mathbb{Z}$ is **recurrent**.
- If $p \neq 1/2$, $4p(1-p) < 1$. The series $\sum \frac{(4p(1-p))^n}{\sqrt{\pi n}}$ converges, so the walk on $\mathbb{Z}$ is **transient**.

### 5.4.2 The symmetric walk on $\mathbb{Z}^2$

Let $(X_n)$ be a sequence of i.i.d. random variables taking values in $\{(1,0), (-1,0), (0,1), (0,-1)\}$, with a probability of $1/4$ for each one. It can be shown that:

$$P(S_{2n} = 0) \sim \frac{1}{\pi n}$$

The series $\sum \frac{1}{n}$ (harmonic series) diverges. The symmetric walk on $\mathbb{Z}^2$ is **recurrent**.

### 5.4.3 The symmetric walk on $\mathbb{Z}^d, d \geq 3$

For the symmetric walk on $\mathbb{Z}^3$, we have:

$$P(S_{2n} = 0) = O(n^{-3/2})$$

The series $\sum n^{-3/2}$ converges (Riemann series with $p = 3/2 > 1$). The symmetric random walk on $\mathbb{Z}^3$ is **transient**. In dimension $d > 3$, the symmetric random walk in $\mathbb{Z}^d$ is transient, and the mean number of returns to 0 is:

$$U_d = \frac{1}{(2\pi)^d} \int_{[-\pi,\pi]^d} \frac{1}{1 - \frac{1}{d}\sum_{k=1}^{d} \cos u_k} du_1 \dots du_d$$

This integral is finite for $d \geq 3$.

# CHAPTER 6

# THE PROBLEM OF RANDOM SIGNS

## 6.1 The problem

Let $(u_n)_{n\in\mathbb{N}}$ be a (deterministic) sequence of real numbers. Let $(\epsilon_n)_{n\in\mathbb{N}}$ be an infinite i.i.d. sequence of Bernoulli(1/2) variables such that:

$$\forall n \in \mathbb{N}, \quad \mathbb{P}(\epsilon_n = -1) = \mathbb{P}(\epsilon_n = 1) = 1/2$$

Does the series $\sum \epsilon_n u_n$ converge?

- If $u_n \not\to 0$, the series does not converge.
- If the series $\sum |u_n|$ is absolutely convergent, then $\sum \epsilon_n u_n$ is almost surely absolutely convergent.

Let's look at the harmonic series: $\sum 1/n = +\infty$.

$$\sum_{n\geq 1} \frac{(-1)^{n-1}}{n} = 1 - 1/2 + 1/3 - \cdots = \ln(2)$$

## 6.2 Convergence in $L^2$

**Proposition 6.2.1.** The series of random variables $\sum u_n \epsilon_n$ converges in $L^2$ if and only if the series $\sum |u_n|^2$ converges.

**Proof.** Let $S_n = \sum_{k=0}^{n} \epsilon_k u_k$. Then, the series $\sum \epsilon_n u_n$ converges in $L^2$ iff the sequence $(S_n)_{n\in\mathbb{N}}$ converges in $L^2$. The space $L^2(\Omega, \mathcal{F}, \mathbb{P})$ is complete with respect to $\|\cdot\|_2$. Thus, the sequence $(S_n)_{n\in\mathbb{N}}$ converges

iff it satisfies the Cauchy criterion. Let $m < n$, consider:

$$\begin{aligned}
\|S_n - S_m\|_2^2 &= \int |S_n - S_m|^2 d\mathbb{P} \\
&= E[(S_n - S_m)^2] \\
&= E\left[\left(\sum_{k=m+1}^{n} \epsilon_k u_k\right)^2\right] \\
&= E\left[\sum_{m<k,l\leq n} \epsilon_k \epsilon_l u_k u_l\right] = \sum_{m<k,l\leq n} u_k u_l E(\epsilon_k \epsilon_l)
\end{aligned}$$

For $k \neq l$, $E(\epsilon_k \epsilon_l) = E(\epsilon_k)E(\epsilon_l) = 0$ because the random variables are independent and centered. For $k = l$, $E(\epsilon_k^2) = 1$. Therefore,

$$\|S_n - S_m\|_2^2 = \sum_{k=m+1}^{n} u_k^2 = a_n - a_m \quad \text{where } a_n = \sum_{k=0}^{n} u_k^2$$

Thus, the sequence $(S_n)$ is Cauchy in $L^2$ iff the sequence $(a_n)$ is Cauchy in $\mathbb{R}$. □

## 6.3 Kolmogorov's 0-1 Law

**Definition 6.3.1.** Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of random variables defined on the space $(\Omega, \mathcal{F}, \mathbb{P})$. The **tail sigma-algebra**, or **asymptotic sigma-algebra**, associated with the sequence $(X_n)$ is the sigma-algebra:

$$\mathcal{T} = \bigcap_{n\geq 0} \sigma(X_n, X_{n+1}, \dots)$$

An event $A \in \mathcal{F}$ is said to be **asymptotic** for the sequence $(X_n)$ if it belongs to the asymptotic sigma-algebra.

**Example 6.3.2.** The following events are asymptotic:

- $\{\limsup X_n > 100\}$
- $\{\liminf X_n \in [0, 1]\}$
- $\{\omega : \lim X_n(\omega) \text{ exists}\}$
- $\{\omega : \sum \epsilon_n(\omega) u_n \text{ converges}\}$, which is an asymptotic event for the sequence $(\epsilon_n)_{n\in\mathbb{N}}$.

**Theorem 6.3.3** (Kolmogorov's 0-1 Law)**.** Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of independent random variables. Let A be an asymptotic event for the sequence $(X_n)$. Then $\mathbb{P}(A)$ is equal to 0 or 1.

### Application to the problem of signs

The event $\{\sum \epsilon_n u_n \text{ converges}\}$ is an asymptotic event for the sequence $(\epsilon_n)$. Since the $(\epsilon_n)$ are independent, the probability of this event is 0 or 1. Either the series converges almost surely or it does not converge almost surely.

## 6.4 An approximation result

The **symmetric difference** of 2 sets A and B is $A\Delta B = (A \cup B) \setminus (A \cap B)$.

- $A\Delta\emptyset = \emptyset\Delta A = A$
- $A\Delta A = \emptyset$
- $\forall A, B, C, \quad A\Delta(B\Delta C) = (A\Delta B)\Delta C$
- $(\mathcal{P}(\Omega), \Delta)$ is an abelian group.

**Proposition 6.4.1.** Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of random variables and let A be an event in the sigma-algebra $\sigma(X_0, X_1, \dots)$. Then there exists a sequence of events $(A_n)_{n\in\mathbb{N}}$ such that:

- $\forall n \in \mathbb{N}, \exists k(n), \quad A_n \in \sigma(X_0, \dots, X_{k(n)})$
- $\lim_{n\to\infty} \mathbb{P}(A_n\Delta A) = 0$

**Remark 6.4.2.** $\forall A, B, \quad |\mathbb{P}(A) - \mathbb{P}(B)| \leq \mathbb{P}(A\Delta B)$, so we also have $\lim_{n\to\infty} \mathbb{P}(A_n) = \mathbb{P}(A)$.

**Proof.** Consider $\mathcal{G}$ the collection of events for which the result holds true. We show that $\mathcal{G}$ is a $\sigma$-algebra. The algebra $\mathcal{A} = \bigcup_m \sigma(X_0, \dots, X_m)$ is contained in $\mathcal{G}$. Indeed, if $A \in \sigma(X_0, \dots, X_m)$, it suffices to take $A_n = A$ for all $n$. We then show that $\mathcal{G}$ is a $\sigma$-algebra.

- $\Omega \in \mathcal{G}$.
- If $A \in \mathcal{G}$ and $(A_n)$ is a sequence approximating $A$, then the sequence $(A_n^c)$ approximates $A^c$:

$$\mathbb{P}(A^c\Delta A_n^c) = \mathbb{P}(A\Delta A_n) \to 0$$

- If $(A_m)_{m\geq 1}$ is a sequence in $\mathcal{G}$, then $A = \bigcup_m A_m \in \mathcal{G}$. Since $A_m \in \mathcal{G}$, there exists an approximating sequence $(A_{m,n})_n$ for $A_m$. $\forall n, \forall m \geq 1, \exists k(m,n), A_{m,n} \in \sigma(X_0, \dots, X_{k(m,n)})$ and $\lim_{n\to\infty} \mathbb{P}(A_m\Delta A_{m,n}) = 0$. Since $\lim_{M\to\infty} \mathbb{P}(A \setminus \bigcup_{m=1}^M A_m) = 0$, $\forall M \geq 1, \exists \varphi(M)$ such that $\mathbb{P}(A \setminus \bigcup_{m=1}^{\varphi(M)} A_m) < 1/M$. Then, $\forall m \in \{1, \dots, \varphi(M)\}$, $\exists N(M,m)$ such that $\forall n > N(M,m)$, $\mathbb{P}(A_m\Delta A_{m,n}) \leq \frac{1}{\varphi(M)M}$. Thus, take $N(M) = \max\{N(M,m), 1 \leq m \leq \varphi(M)\}$ and $A'_M = \bigcup_{m=1}^{\varphi(M)} A_{m,N(M,m)}$.

$$\begin{aligned}
\mathbb{P}(A\Delta A'_M) &\leq \mathbb{P}\left(A\Delta \bigcup_{m=1}^{\varphi(M)} A_m\right) + \mathbb{P}\left(\left(\bigcup_{m=1}^{\varphi(M)} A_m\right)\Delta A'_M\right) \\
&< \frac{1}{M} + \sum_{m=1}^{\varphi(M)} \mathbb{P}(A_m\Delta A_{m,N(M,m)}) \\
&\leq \frac{1}{M} + \sum_{m=1}^{\varphi(M)} \frac{1}{\varphi(M)M} = \frac{1}{M} + \frac{1}{M} = \frac{2}{M} \to 0
\end{aligned}$$

Since $\mathcal{G}$ is a $\sigma$-algebra containing $\mathcal{A}$, $\mathcal{G}$ contains $\sigma(\mathcal{A}) = \sigma(X_0, X_1, \dots)$. □

We apply this result to prove the 0-1 law. Let $(X_n)$ be independent, $A \in \mathcal{T} = \bigcap_k \sigma(X_k, X_{k+1}, \dots)$. There exist $(A_n)$ such that $A_n \in \sigma(X_0, \dots, X_{k(n)})$ and $\lim \mathbb{P}(A\Delta A_n) = 0$. First, since $|\mathbb{P}(A) - \mathbb{P}(A_n)| \leq \mathbb{P}(A\Delta A_n) \to 0$, we have $\mathbb{P}(A_n) \to \mathbb{P}(A)$. Furthermore, $A_n \in \sigma(X_0, \dots, X_{k(n)})$ and $A \in \mathcal{T} \subset \sigma(X_{k(n)+1}, X_{k(n)+2}, \dots)$.

Because $(X_n)$ are independent, the two *sigma*-algebras are independent. Thus, $A$ and $A_n$ are also independent.

$$\mathbb{P}(A \cap A_n) = \mathbb{P}(A)\mathbb{P}(A_n) \xrightarrow{n\to\infty} \mathbb{P}(A)^2$$

Similarly, $|\mathbb{P}(A) - \mathbb{P}(A \cap A_n)| = \mathbb{P}(A \setminus A_n) \leq \mathbb{P}(A \Delta A_n) \to 0$, hence $\lim \mathbb{P}(A \cap A_n) = \mathbb{P}(A)$. We conclude that $\mathbb{P}(A) = \mathbb{P}(A)^2$, which implies $\mathbb{P}(A) \in \{0, 1\}$.

## 6.5 Kolmogorov's Inequality

**Theorem 6.5.1.** Let $X_1, \ldots, X_n$ be n independent, centered random variables ($E(X_i) = 0$) with finite variances. Let $\forall k \in \{1, \ldots, n\}, S_k = X_1 + \cdots + X_k$. Then $\forall \alpha > 0$,

$$\mathbb{P}(\max_{1\leq k\leq n} |S_k| \geq \alpha) \leq \frac{1}{\alpha^2}\mathrm{Var}(S_n)$$

**Proof.** For $k \in \{1, \ldots, n\}$, we introduce the event $A_k = \{|S_1| < \alpha, \ldots, |S_{k-1}| < \alpha, |S_k| \geq \alpha\}$. The events $(A_k)$ are disjoint. Let $A = \bigcup_{k=1}^n A_k = \{\max_{1\leq k\leq n} |S_k| \geq \alpha\}$.

$$\begin{aligned}
\mathrm{Var}(S_n) = E(S_n^2) &= \int_\Omega S_n^2 d\mathbb{P} \geq \int_A S_n^2 d\mathbb{P} \\
&= \sum_{k=1}^n \int_{A_k} S_n^2 d\mathbb{P} \quad \text{since the } A_k \text{ are disjoint}
\end{aligned}$$

Furthermore,

$$\begin{aligned}
\int_{A_k} S_n^2 d\mathbb{P} &= \int_{A_k} (S_k + (S_n - S_k))^2 d\mathbb{P} \\
&= \int_{A_k} \left(S_k^2 + 2S_k(S_n - S_k) + (S_n - S_k)^2\right) d\mathbb{P} \\
&\geq \int_{A_k} \left(S_k^2 + 2S_k(S_n - S_k)\right) d\mathbb{P}
\end{aligned}$$

The term $\mathbf{1}_{A_k} S_k$ is $\sigma(X_1, \ldots, X_k)$-measurable. The term $S_n - S_k = \sum_{j=k+1}^n X_j$ is $\sigma(X_{k+1}, \ldots, X_n)$-measurable. By independence,

$$E[\mathbf{1}_{A_k} S_k(S_n - S_k)] = E[\mathbf{1}_{A_k} S_k]E[S_n - S_k] = E[\mathbf{1}_{A_k} S_k] \cdot 0 = 0$$

Therefore $\int_{A_k} 2S_k(S_n - S_k)d\mathbb{P} = 0$. We thus have $\int_{A_k} S_n^2 d\mathbb{P} \geq \int_{A_k} S_k^2 d\mathbb{P}$. On $A_k$, we have $|S_k| \geq \alpha$, so $S_k^2 \geq \alpha^2$.

$$\int_{A_k} S_k^2 d\mathbb{P} \geq \int_{A_k} \alpha^2 d\mathbb{P} = \alpha^2 \mathbb{P}(A_k)$$

Summing over $k$:

$$\mathrm{Var}(S_n) \geq \sum_{k=1}^n \alpha^2 \mathbb{P}(A_k) = \alpha^2 \mathbb{P}\left(\bigcup_{k=1}^n A_k\right) = \alpha^2 \mathbb{P}(A)$$

Which gives the result. □

## 6.6 Convergence of series of independent random variables

**Theorem 6.6.1.** Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of independent, centered random variables with finite variances $\forall n, E(X_n) = 0, E(X_n^2) < \infty$. If $\sum E(X_n^2) < \infty$, then the series $\sum X_n$ converges almost surely.

A consequence of the previous results is that, for the random sign problem:

$$\sum_{n\in\mathbb{N}} u_n^2 < \infty \iff \sum_{n\in\mathbb{N}} \epsilon_n u_n \text{ converges a.s.}$$

# CHAPTER 7

# THE LAWS OF LARGE NUMBERS

## 7.1 Fundamental inequalities

### 7.1.1 Markov's Inequality

**Proposition 7.1.1.** Let $X$ be a positive random variable. Then $\forall \lambda > 0$,

$$P(X \geq \lambda) \leq \frac{1}{\lambda} E[X].$$

**Remark 7.1.2.** Since $X \geq 0$, $E[X]$ is always defined, possibly with $E[X] = +\infty$.

**Proof.** We have

$$\begin{aligned} E[X] = \int_\Omega X dP &= \int_{\{X \geq \lambda\}} X dP + \int_{\{X < \lambda\}} X dP \\ &\geq \int_{\{X \geq \lambda\}} X dP \quad \text{(since } X \geq 0) \\ &\geq \int_{\{X \geq \lambda\}} \lambda dP \quad \text{(because on } \{X \geq \lambda\}, X \geq \lambda) \\ &= \lambda P(X \geq \lambda). \end{aligned}$$

By dividing by $\lambda > 0$, we obtain the result. □

Markov's inequality is used to prove many more general inequalities.

### 7.1.2 Chebyshev's Inequality

**Proposition 7.1.3.** Let $X$ be a random variable with a finite second moment, i.e., $E[X^2] < +\infty$. Then $\forall t > 0$,

$$P(|X - E[X]| \geq t) \leq \frac{1}{t^2} \mathrm{Var}(X).$$

**Proof.** The inequality is trivial if $t \leq 0$. Let $t > 0$. We have

$$P(|X - E[X]| \geq t) = P((X - E[X])^2 \geq t^2).$$

We apply Markov's inequality to the positive random variable $Y = (X - E[X])^2$ with $\lambda = t^2$:

$$P(Y \geq t^2) \leq \frac{E[Y]}{t^2}.$$

Now, $E[Y] = E[(X - E[X])^2] = \mathrm{Var}(X)$. Hence the result:

$$P(|X - E[X]| \geq t) \leq \frac{\mathrm{Var}(X)}{t^2}.$$

□

The value of this inequality lies in its simplicity and the fact that it is very well suited to sums of random variables.

## 7.2 The weak law of large numbers

**Theorem 7.2.1.** Let $\mu$ be a probability measure on $\mathbb{R}$ that is integrable, i.e., $\int_{\mathbb{R}} |x| d\mu(x) < +\infty$. Let $m = \int_{\mathbb{R}} x d\mu(x)$. For $n \geq 1$, let $X_1, \ldots, X_n$ be i.i.d. random variables with distribution $\mu$. Then, for $\forall \varepsilon > 0$,

$$\lim_{n \to \infty} P\left(\left|\frac{X_1 + \cdots + X_n}{n} - m\right| \geq \varepsilon\right) = 0.$$

**Remark 7.2.2.** If $(X_n)$ is a sequence of i.i.d. random variables with distribution $\mu$, this can be written as:

$$\frac{X_1 + \cdots + X_n}{n} \xrightarrow[n \to \infty]{P} m.$$

**Remark 7.2.3.** In fact, by applying the transfer formula to the vector $(X_1, \ldots, X_n)$, we have:

$$P\left(\left|\frac{X_1 + \cdots + X_n}{n} - m\right| \geq \varepsilon\right) = P_{\mu^{\otimes n}}\left(\{(x_1, \ldots, x_n) \in \mathbb{R}^n, \left|\frac{x_1 + \cdots + x_n}{n} - m\right| \geq \varepsilon\}\right).$$

Thus, the weak law is actually an asymptotic result on the sequence of laws $(\mu^{\otimes n}, n \geq 1)$. The law does not require having an i.i.d. sequence.

**Proof** (Proof (under the assumption of a second-order moment))**.** Suppose that the distribution $\mu$ has a second-order moment. Let $\sigma^2 = \mathrm{Var}(\mu) = \int_{\mathbb{R}} (x - m)^2 d\mu(x)$. Let $\bar{X}_n = \frac{X_1 + \cdots + X_n}{n}$. The expectation calculation gives:

$$E[\bar{X}_n] = E\left[\frac{X_1 + \cdots + X_n}{n}\right] = \frac{1}{n}(E[X_1] + \cdots + E[X_n]) = \frac{1}{n}(n \cdot m) = m.$$

Let $\varepsilon > 0$. By applying the Bienaymé-Chebyshev inequality to $\bar{X}_n$:

$$P(|\bar{X}_n - m| \geq \varepsilon) \leq \frac{\mathrm{Var}(\bar{X}_n)}{\varepsilon^2}.$$

We calculate the variance:

$$\begin{aligned}
\mathrm{Var}(\bar{X}_n) &= \mathrm{Var}\left(\frac{X_1 + \cdots + X_n}{n}\right) \\
&= \frac{1}{n^2}\mathrm{Var}(X_1 + \cdots + X_n) \\
&= \frac{1}{n^2}(\mathrm{Var}(X_1) + \cdots + \mathrm{Var}(X_n)) \quad \text{(par indépendance)} \\
&= \frac{1}{n^2}(n \cdot \sigma^2) \quad \text{(car les lois sont identiques)} \\
&= \frac{\sigma^2}{n}.
\end{aligned}$$

Thus,

$$P\left(\left|\frac{X_1 + \cdots + X_n}{n} - m\right| \geq \varepsilon\right) \leq \frac{\sigma^2}{n\varepsilon^2} \xrightarrow[n\to\infty]{} 0.$$

□

## 7.3 The strong law of large numbers

**Theorem 7.3.1** (Strong Law of Large Numbers)**.** Let $(X_n)_{n\geq 1}$ be a sequence of i.i.d. integrable random variables, that is, $E[|X_1|] < +\infty$. Then

$$\frac{X_1 + \cdots + X_n}{n} \xrightarrow[n\to\infty]{p.s.} E[X_1].$$

**Proof** (Proof Sketch)**.** The proof is divided into several steps.

**Step 1: Centering**

Let $m = E[X_1]$. By replacing $X_n$ with $X_n - m$ for $n \geq 1$, we can assume that the sequence is centered, meaning that $E[X_i] = 0$ for all $i$.

**Step 2: Truncation**

Let, for all $n \geq 1$, $Y_n = X_n \mathbb{I}_{\{|X_n| \leq n\}}$. That is:

$$Y_n = \begin{cases} X_n & \text{si } |X_n| \leq n \\ 0 & \text{sinon} \end{cases}$$

We can then decompose the average as follows:

$$\frac{X_1 + \cdots + X_n}{n} = \frac{(X_1 - Y_1) + \cdots + (X_n - Y_n)}{n} + \frac{Y_1 + \cdots + Y_n}{n}.$$

The proof consists of showing that the two terms on the right-hand side tend to 0 almost surely. For the first term, we use the following lemma. □

**Lemma 7.3.2.** Let $(X_n)_{n\geq 1}$ be a sequence of i.i.d. random variables. The following are equivalent:

i) $E[|X_1|] < +\infty$

ii) $P(|X_n| > n \text{ a.s.}) = 0$

**Proof.** Since the events $\{|X_n| > n\}$ for $n \geq 1$ are not independent (as they share the same distribution), we cannot directly apply the second Borel-Cantelli lemma. However, the first Borel-Cantelli lemma applies. The events $\{|X_n| > n\}$ are independent. By the Borel-Cantelli lemma, we have the equivalence:

$$P(|X_n| > n \text{ i.s.}) = 0 \iff \sum_{n\geq 1} P(|X_n| > n) < +\infty.$$

Since the variables $X_n$ are identically distributed, $P(|X_n| > n) = P(|X_1| > n)$, so the equivalence becomes

$$\sum_{n\geq 1} P(|X_1| > n) < +\infty.$$

It remains to show that $\sum_{n\geq 1} P(|X_1| > n) < +\infty \iff E[|X_1|] < +\infty$. For a positive random variable $Z$, we have $E[Z] = \int_0^{+\infty} P(Z > t)dt$. Applying this to $Z = |X_1|$:

$$\begin{aligned} E[|X_1|] &= \int_0^{+\infty} P(|X_1| > t)dt \\ &= \sum_{n=0}^{\infty} \int_n^{n+1} P(|X_1| > t)dt. \end{aligned}$$

For $t \in [n, n+1)$, we have the bound $P(|X_1| > n+1) \leq P(|X_1| > t) \leq P(|X_1| > n)$. Integrating over $[n, n+1]$, we obtain:

$$P(|X_1| > n+1) \leq \int_n^{n+1} P(|X_1| > t)dt \leq P(|X_1| > n).$$

Summing over $n \geq 0$:

$$\sum_{n=0}^{\infty} P(|X_1| > n+1) \leq \sum_{n=0}^{\infty} \int_n^{n+1} P(|X_1| > t)dt \leq \sum_{n=0}^{\infty} P(|X_1| > n).$$

This can be rewritten as:

$$\sum_{k=1}^{\infty} P(|X_1| > k) \leq E[|X_1|] \leq \sum_{k=0}^{\infty} P(|X_1| > k) = P(|X_1| > 0) + \sum_{k=1}^{\infty} P(|X_1| > k).$$

The expectation $E[|X_1|]$ is therefore finite if and only if the series $\sum_{n\geq 1} P(|X_1| > n)$ converges. □

By the lemma, since $E[|X_1|] < \infty$, we have $P(|X_n| > n \text{ a.s.}) = 0$, which is equivalent to $P(X_n \neq Y_n \text{ a.s.}) = 0$. This implies that the series $\sum(X_n - Y_n)$ converges almost surely. From Kronecker's lemma, it follows that

$$\frac{(X_1 - Y_1) + \cdots + (X_n - Y_n)}{n} \xrightarrow[n\to\infty]{p.s.} 0.$$

The complete proof of the strong law of large numbers then requires showing that the second term $\frac{1}{n}\sum_{i=1}^{n} Y_i$ also converges to 0 a.s., which is done using Kolmogorov's criterion.

## 7.4 Kolmogorov Criterion

**Proposition 7.4.1.** Let $(Z_n)_{n\geq 1}$ be a sequence of independent and centered random variables ($E[Z_n] = 0$ for all $n$). If $\sum_{n\geq 1} \frac{E[Z_n^2]}{n^2} < +\infty$, then the series $\sum_{n\geq 1} \frac{Z_n}{n}$ converges almost surely, and

$$\frac{Z_1 + \cdots + Z_n}{n} \xrightarrow[n\to\infty]{p.s.} 0.$$

**Proof.** The proof is omitted. □

# CHAPTER 8

# CONVERGENCE IN LAW

The law is the most important thing.

## 8.1 Definition

There exist sequences of random variables that converge neither almost surely, nor in probability, nor in $L^1$, yet whose distribution converges, or may even be constant.

**Example 8.1.1.** Consider the probability space $(\Omega, \mathcal{F}, P) = ([0,1], \mathcal{B}([0,1]), \lambda)$ where $\lambda$ is the Lebesgue measure. Consider the sequence of random variables $(X_n)$ defined for $\omega \in [0,1]$ by:

$$X_{2n}(\omega) = \mathbf{1}_{[0,1/2]}(\omega)$$
$$X_{2n+1}(\omega) = \mathbf{1}_{[1/2,1]}(\omega)$$

This sequence converges neither almost surely, nor in $L^1$, nor in probability, but all the random variables follow the Bernoulli distribution with parameter $1/2$.

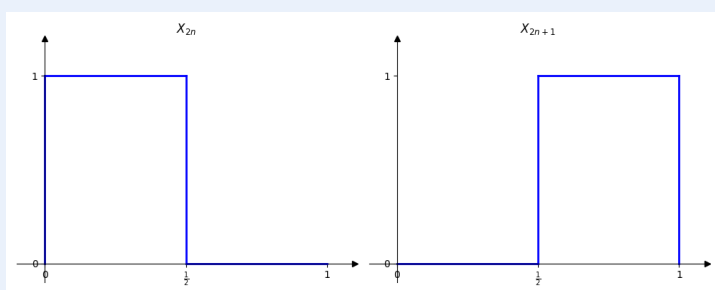



Figure 8.1: Representation of the random variables $X_{2n}$ and $X_{2n+1}$.

For every $n$, the law of $X_n$ is the Bernoulli(1/2) distribution.

$$\forall n, P(X_n = 0) = P(X_n = 1) = \frac{1}{2}$$

and thus $P_{X_n} = \text{Ber}(1/2)$ does not depend on $n$.

**Definition 8.1.2** (Convergence in Distribution)**.** Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of real random variables (not necessarily defined on the same probability space). The sequence $(X_n)_{n\in\mathbb{N}}$ converges in law to a random variable $X$, denoted $X_n \xrightarrow{\mathcal{L}} X$, if and only if, for every bounded continuous function $f : \mathbb{R} \to \mathbb{R}$,

$$\lim_{n\to\infty} \mathbb{E}[f(X_n)] = \mathbb{E}[f(X)].$$

**Remark 8.1.3.** The expectation $\mathbb{E}[f(X_n)]$ depends only on the law of $X_n$, i.e.,

$$\mathbb{E}[f(X_n)] = \int_\Omega f(X_n)dP = \int_{\mathbb{R}} f(x)dP_{X_n}(x).$$

Thus, convergence in law applies even in situations where the random variables $X_n$ are not defined on the same probability space $(\Omega, \mathcal{F}, P)$, i.e., if $X_n : (\Omega_n, \mathcal{F}_n, P_n) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, where the $(\Omega_n, \mathcal{F}_n, P_n)$ have no a priori relation. This fundamentally distinguishes convergence in law from other modes of convergence.

## 8.2 The cumulative distribution function

A fundamental tool for studying probability laws on $\mathbb{R}$ is the cumulative distribution function.

**Definition 8.2.1.** Let $X$ be a real-valued random variable defined on $(\Omega, \mathcal{F}, P)$. Its cumulative distribution function is the function $F_X$ defined by:

$$\forall x \in \mathbb{R}, \quad F_X(x) = P(X \leq x).$$

We have $F_X(x) = P_X((-\infty, x])$. Since the intervals $(-\infty, x]$, $x \in \mathbb{R}$ generate the Borel sets and form a family stable under finite intersections, by the uniqueness of the $\pi$-$\pi - \lambda$ extension theorem, the law $P_X$ is determined by $F_X$. Conclusion: the cumulative distribution function $F_X$ determines the law $P_X$ of $X$.

A cumulative distribution function $F$ satisfies:

i) $F$ is increasing from $\mathbb{R}$ to $[0, 1]$.

ii) $F$ is right-continuous at every point.

iii) $\lim_{x\to-\infty} F(x) = 0$ and $\lim_{x\to+\infty} F(x) = 1$.

**Proof.** i) By the increasing nature of the measure. If $x_1 \leq x_2$, then $(-\infty, x_1] \subseteq (-\infty, x_2]$, so $P(X \leq x_1) \leq P(X \leq x_2)$.

ii) Let $x_0 \in \mathbb{R}$. Consider $\lim_{x\to x_0, x>x_0} F_X(x)$. Let $(x_n)_{n\in\mathbb{N}}$ be a sequence decreasing towards $x_0$. Then the sets $A_n = (-\infty, x_n]$ form a decreasing sequence of events. By monotone continuity of probability:

$$\lim_{n\to\infty} F_X(x_n) = \lim_{n\to\infty} P(X \in A_n) = P(X \in \bigcap_{n=1}^{\infty} A_n) = P(X \in (-\infty, x_0]) = F_X(x_0).$$

iii) By monotone continuity of probability. For the limit at $-\infty$, we take a sequence $x_n \to -\infty$ (e.g., $x_n = -n$).

$$\lim_{x\to-\infty} F_X(x) = \lim_{n\to\infty} P(X \leq -n) = P(\bigcap_{n=1}^{\infty} \{X \leq -n\}) = P(\emptyset) = 0.$$

For the limit at $+\infty$, we take a sequence $x_n \to +\infty$ (e.g., $x_n = n$).

$$\lim_{x\to+\infty} F_X(x) = \lim_{n\to\infty} P(X \le n) = P(\bigcup_{n=1}^{\infty}\{X \le n\}) = P(X \in \mathbb{R}) = 1.$$

□

**Remark 8.2.2.** In some books, $F_X$ is defined as $F_X(x) = P(X < x)$. In this case, the distribution function is left-continuous.

**Example 8.2.3** (Cumulative Distribution Functions of Standard Distributions).

1. $X \sim \mathcal{B}(p)$ (Bernoulli distribution). $F_X(x) = 0$ if $x < 0$, $1-p$ if $0 \le x < 1$, and 1 if $x \ge 1$.
2. $X \sim \mathcal{U}(\{1,\dots,n\})$ (Discrete uniform distribution). $F_X(x) = \frac{\lfloor x \rfloor}{n}$ for $1 \le x \le n$.
3. $X \sim \mathcal{U}([0,1])$ (Continuous uniform distribution). $F_X(x) = 0$ if $x < 0$, $x$ if $0 \le x \le 1$, and 1 if $x > 1$.
4. $X \sim \mathrm{Exp}(\lambda)$ (Exponential distribution). $F_X(x) = 1 - e^{-\lambda x}$ if $x \ge 0$, and 0 if $x < 0$.
5. $X \sim \mathcal{N}(0,1)$ (Standard normal distribution). $F_X(x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$.

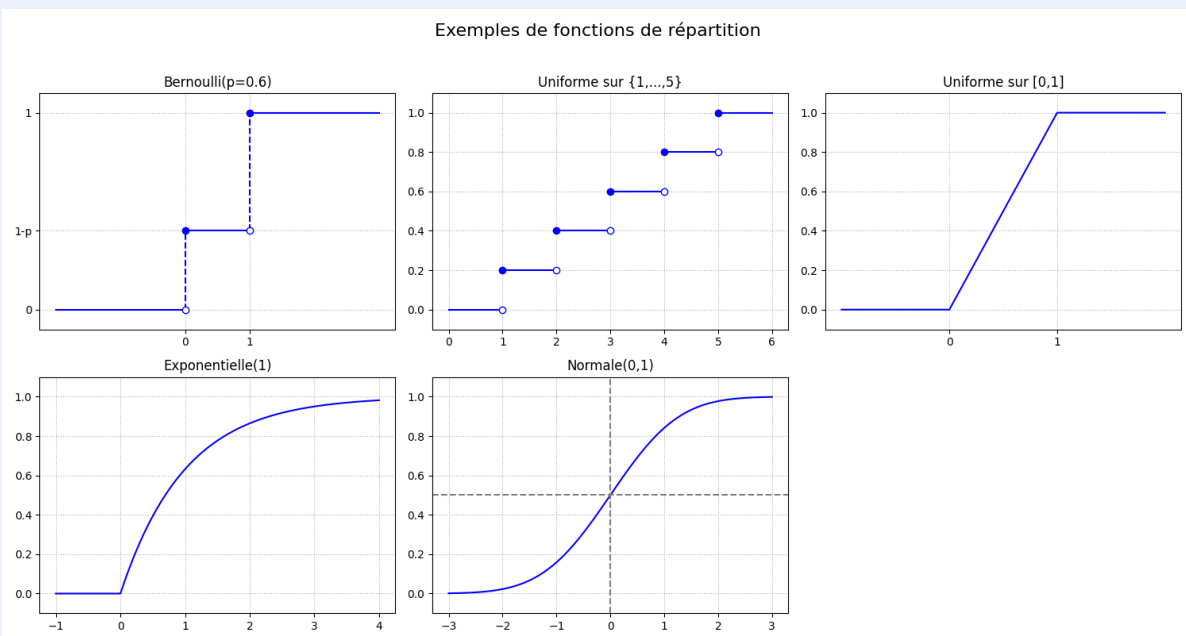



Figure 8.2: Examples of cumulative distribution functions.

More generally, if the law of $X$ has a density $f$, then its cumulative distribution function is given by

$$F_X(x) = \int_{-\infty}^{x} f(t)dt.$$

**Theorem 8.2.4.** Let $\mathcal{M}_1(\mathbb{R})$ denote the set of probability measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, and let $\mathfrak{R}$ denote the set of functions satisfying the three properties of a distribution function. The mapping $\mu \mapsto F_\mu$, where $F_\mu(x) = \mu((-\infty, x])$, is a bijection from $\mathcal{M}_1(\mathbb{R})$ to $\mathfrak{R}$.

This is a difficult result, which in particular allows for the construction of singular probability measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

**Example 8.2.5** (Construction of a Singular Law)**.** One can construct a sequence of distribution functions $(F_n)$ on $[0, 1]$ that converges uniformly to a continuous function $F_\infty$, the Cantor function. This function $F_\infty$ is a distribution function, but the associated probability measure $\mu$ is singular with respect to the Lebesgue measure $\lambda$. It has no atoms ($\mu(\{x\}) = 0$ for all $x$) and is concentrated on a set of Lebesgue measure zero (the Cantor set).

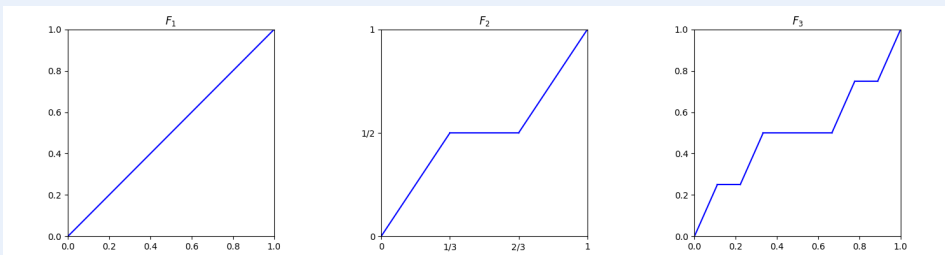



Figure 8.3: First steps in the construction of the Cantor function.

## 8.3 Convergence in law and cumulative distribution functions

Since convergence in law deals with the laws of random variables, it should be expressible via distribution functions.

**Theorem 8.3.1.** Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of real random variables and $X$ another real random variable. Let $(F_n)_{n\in\mathbb{N}}$ and $F$ be their corresponding cumulative distribution functions. The following statements are equivalent:

i) The sequence $(X_n)$ converges in law to $X$ ($X_n \xrightarrow{\mathcal{L}} X$).

ii) At every point $x \in \mathbb{R}$ where $F$ is continuous, the sequence $(F_n(x))$ converges to $F(x)$, i.e., $\lim_{n\to\infty} F_n(x) = F(x)$.

**Proof.** $(i) \Rightarrow (ii)$: Let $x_0$ be a point where $F$ is continuous. We need to show that $\lim_{n\to\infty} F_n(x_0) = F(x_0)$. For any $\epsilon > 0$, we can construct two bounded continuous functions $f_\epsilon$ and $g_\epsilon$ that "bound" the indicator function $\mathbf{1}_{(-\infty,x_0]}$ as follows:

$$f_\epsilon \leq \mathbf{1}_{(-\infty,x_0]} \leq g_\epsilon$$

such that $\mathbb{E}[g_\epsilon(X)] - \mathbb{E}[f_\epsilon(X)]$ is small. For example, let $f_\epsilon$ be equal to 1 on $(-\infty, x_0 - \epsilon]$, 0 on $[x_0, \infty)$ and linear in between. Let $g_\epsilon$ be equal to 1 on $(-\infty, x_0]$, 0 on $[x_0 + \epsilon, \infty)$ and linear in between.

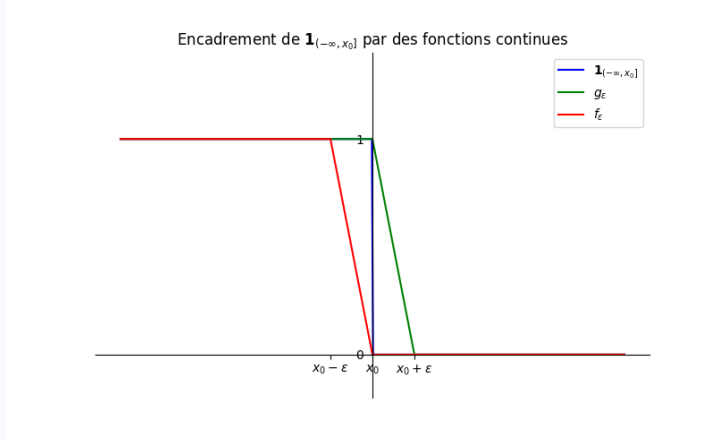



Figure 8.4: Functions $f_\epsilon$ and $g_\epsilon$ bounding $\mathbf{1}_{(-\infty,x_0]}$.

We have $f_\epsilon(y) \leq \mathbf{1}_{(-\infty,x_0]}(y) \leq g_\epsilon(y)$ for all $y \in \mathbb{R}$. Taking the expectation with respect to $X_n$:

$$\mathbb{E}[f_\epsilon(X_n)] \leq \mathbb{E}[\mathbf{1}_{(-\infty,x_0]}(X_n)] \leq \mathbb{E}[g_\epsilon(X_n)]$$

$$\mathbb{E}[f_\epsilon(X_n)] \leq F_n(x_0) \leq \mathbb{E}[g_\epsilon(X_n)]$$

Since $f_\epsilon$ and $g_\epsilon$ are continuous and bounded, and since $X_n \xrightarrow{\mathcal{L}} X$, we have:

$$\lim_{n\to\infty} \mathbb{E}[f_\epsilon(X_n)] = \mathbb{E}[f_\epsilon(X)]$$
$$\lim_{n\to\infty} \mathbb{E}[g_\epsilon(X_n)] = \mathbb{E}[g_\epsilon(X)]$$

Taking the limit in the inequality, we obtain:

$$\mathbb{E}[f_\epsilon(X)] \leq \liminf_{n\to\infty} F_n(x_0) \leq \limsup_{n\to\infty} F_n(x_0) \leq \mathbb{E}[g_\epsilon(X)]$$

Moreover, $\mathbb{E}[f_\epsilon(X)] \geq P(X \leq x_0 - \epsilon) = F(x_0 - \epsilon)$ and $\mathbb{E}[g_\epsilon(X)] \leq P(X \leq x_0 + \epsilon) = F(x_0 + \epsilon)$. Thus,

$$F(x_0 - \epsilon) \leq \liminf_{n\to\infty} F_n(x_0) \leq \limsup_{n\to\infty} F_n(x_0) \leq F(x_0 + \epsilon)$$

Since $x_0$ is a continuity point of $F$, we have $\lim_{\epsilon\to 0} F(x_0 - \epsilon) = \lim_{\epsilon\to 0} F(x_0 + \epsilon) = F(x_0)$. Taking $\epsilon$ to 0, by the squeeze theorem, we obtain:

$$\lim_{n\to\infty} F_n(x_0) = F(x_0).$$

$(ii) \Rightarrow (i)$: Suppose that $F_{X_n}(x) \to F_X(x)$ at every point $x$ where $F_X$ is continuous. Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous and bounded function. We need to show that $E[f(X_n)] \to E[f(X)]$. Let $\epsilon > 0$. Since $\lim_{x\to-\infty} F_X(x) = 0$ and $\lim_{x\to+\infty} F_X(x) = 1$, there exist $a < b$, continuity points of $F_X$, such that $F_X(a) < \epsilon$ and $F_X(b) > 1 - \epsilon$ (hence $P(X > b) < \epsilon$). We decompose the expectation:

$$E[f(X_n)] = \int_{x\leq a} f(x)dP_{X_n} + \int_{a<x\leq b} f(x)dP_{X_n} + \int_{x>b} f(x)dP_{X_n}$$

The integrals over the tails are small. For $n$ large enough, $F_{X_n}(a) \to F_X(a) < \epsilon$ and $1 - F_{X_n}(b) \to 1 - F_X(b) < \epsilon$.

$$\left|\int_{x \le a} f(x) dP_{X_n}\right| \le \|f\|_\infty P(X_n \le a) = \|f\|_\infty F_{X_n}(a) \xrightarrow{n\to\infty} \|f\|_\infty F_X(a) \le \epsilon \|f\|_\infty$$
$$\left|\int_{x > b} f(x) dP_{X_n}\right| \le \|f\|_\infty P(X_n > b) = \|f\|_\infty (1 - F_{X_n}(b)) \xrightarrow{n\to\infty} \|f\|_\infty (1 - F_X(b)) \le \epsilon \|f\|_\infty$$

On the interval $[a, b]$, $f$ is uniformly continuous. It can be approximated by a step function. Let $a = x_0 < x_1 < \cdots < x_k = b$ be a subdivision of continuity points of $F_X$. Define the step function $f_k^+$ by:

$$f_k^+(x) = \sum_{l=1}^{k} \mathbf{1}_{]x_{l-1}, x_l]}(x) \sup_{t \in [x_{l-1}, x_l]} f(t)$$

Then $f(x) \le f_k^+(x)$ on $]a, b]$. We have

$$\int_{a<x\le b} f(x) dP_{X_n} \le \int_{a<x\le b} f_k^+(x) dP_{X_n} = \sum_{l=1}^{k} \left( \sup_{t \in [x_{l-1}, x_l]} f(t) \right) (F_{X_n}(x_l) - F_{X_n}(x_{l-1}))$$

Taking the limit as $n \to \infty$, since the $x_l$ are continuity points:

$$\limsup_{n\to\infty} \int_{a<x\le b} f(x) dP_{X_n} \le \sum_{l=1}^{k} \left( \sup_{t \in [x_{l-1}, x_l]} f(t) \right) (F_X(x_l) - F_X(x_{l-1})) = E[f_k^+(X) \mathbf{1}_{]a,b]}]$$

By uniform continuity of $f$ on $[a, b]$, as the subdivision step tends to 0 ($k \to \infty$), $f_k^+(x) \to f(x)$. By dominated convergence (since $|f_k^+| \le \|f\|_\infty$), we have $E[f_k^+(X) \mathbf{1}_{]a,b]}] \to E[f(X) \mathbf{1}_{]a,b]}]$. Combining the bounds on the tails and the central interval, we obtain for arbitrarily small $\epsilon$ and sufficiently large $n$ and $k$:

$$\limsup_{n\to\infty} E[f(X_n)] \le E[f(X)]$$

A similar argument with a lower bounding step function $f_k^-$ shows that:

$$\liminf_{n\to\infty} E[f(X_n)] \ge E[f(X)]$$

We conclude that $\lim_{n\to\infty} E[f(X_n)] = E[f(X)]$. $\square$

# CHAPTER 9

# CHARACTERISTIC FUNCTIONS

If $f$ is a function from $\mathbb{R}$ to $\mathbb{C}$ and $\mu$ is a measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, we define

$$\int f d\mu = \int \text{Re } f d\mu + i \int \text{Im } f d\mu,$$

when the right-hand side is meaningful, that is, provided that Re $f$ and Im $f$ are $\mu$-integrable.

## 9.1 Definitions

**Definition 9.1.1** (Characteristic Function of a Measure $\mu$)**.** Let $\mu$ be a probability measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Its characteristic function is

$$\forall u \in \mathbb{R} \quad \varphi_\mu(u) = \int_{\mathbb{R}} e^{iux} d\mu(x).$$

This function is well-defined for all $u \in \mathbb{R}$, since the function $x \mapsto e^{iux}$ is continuous and bounded because $|e^{iux}| = 1$. Consequently, its real part ($\cos(ux)$) and imaginary part ($\sin(ux)$) are integrable with respect to any probability measure.

**Definition 9.1.2** (Characteristic Function of a R.V.)**.** Let $X$ be a real random variable defined on $(\Omega, \mathcal{F}, P)$. Its characteristic function is

$$\forall u \in \mathbb{R} \quad \varphi_X(u) = E[e^{iux}].$$

The connection between the two definitions is given by the transfer formula. The expectation of $e^{iux}$ is the integral of the function $x \mapsto e^{iux}$ with respect to the probability distribution $P_X$ of the variable $X$.

$$E[e^{iux}] = \int_{\mathbb{R}} e^{iux} dP_X(x).$$

Thus, the characteristic function of a random variable $X$ is the characteristic function of its distribution $P_X$.

$$\varphi_X = \varphi_{P_X}.$$

The characteristic function is, up to a sign, the Fourier transform of the distribution $P_X$.

## 9.2 Properties

**Proposition 9.2.1.** If $X$ is a real random variable, its characteristic function $\varphi_X$ satisfies:

i) $\varphi_X(0) = 1$.

ii) $\forall u \in \mathbb{R}, |\varphi_X(u)| \leq 1$.

iii) $\forall u \in \mathbb{R}, \varphi_X(-u) = \overline{\varphi_X(u)}$.

iv) $\varphi_X$ is continuous on $\mathbb{R}$.

**Proof.** i) It is obvious: $\varphi_X(0) = E[e^{i \cdot 0 \cdot X}] = E[e^0] = E[1] = 1$.

ii) Using the growth of expectation and the fact that the expectation of a constant is that constant:

$$|\varphi_X(u)| = |E[e^{iux}]| \leq E[|e^{iux}|] = E[1] = 1.$$

iii) By the properties of complex conjugates and the linearity of expectation:

$$\overline{\varphi_X(u)} = \overline{E[e^{iux}]} = E[\overline{e^{iux}}] = E[e^{-iux}] = \varphi_X(-u).$$

iv) To show continuity, let $u_0 \in \mathbb{R}$. We want to show that $\lim_{u \to u_0} \varphi_X(u) = \varphi_X(u_0)$.

$$\varphi_X(u) = \int_\Omega e^{iux} dP.$$

This is an integral depending on a parameter $u$. The function under the integral, $g(u, \omega) = e^{iuX(\omega)}$, is continuous in $u$ for all $\omega$. Moreover, we have a uniform bound:

$$|e^{iux}| \leq 1, \quad \forall u \in \mathbb{R}, \forall x \in \mathbb{R}.$$

The constant function 1 is $P$-integrable. By the dominated convergence theorem, we can interchange the limit and the integral:

$$\lim_{u \to u_0} \varphi_X(u) = \lim_{u \to u_0} \int_\Omega e^{iuX} dP = \int_\Omega \lim_{u \to u_0} e^{iuX} dP = \int_\Omega e^{iu_0 X} dP = \varphi_X(u_0).$$

Therefore, $\varphi_X$ is continuous on $\mathbb{R}$.

□

**Proposition 9.2.2** (Effect of an Affine Transformation)**.** Let $X$ be a random variable and $\alpha, \beta \in \mathbb{R}$. The characteristic function of the random variable $\alpha X + \beta$ is given by:

$$\varphi_{\alpha X + \beta}(u) = e^{iu\beta} \varphi_X(\alpha u).$$

**Proof.** The calculation is straightforward using the properties of expectation:

$$\begin{aligned}
\varphi_{\alpha X + \beta}(u) &= E[e^{iu(\alpha X + \beta)}] \\
&= E[e^{iu\alpha X} e^{iu\beta}] \\
&= e^{iu\beta} E[e^{i(u\alpha)X}] \\
&= e^{iu\beta} \varphi_X(\alpha u).
\end{aligned}$$

□

## 9.3 Classic examples

**Example 9.3.1** (Bernoulli Distribution)**.** If $X \sim Ber(p)$, then $P(X=1)=p$ and $P(X=0)=1-p$.

$$\varphi_X(u) = E[e^{iux}] = P(X=0)e^{iu\cdot 0} + P(X=1)e^{iu\cdot 1} = (1-p)\cdot 1 + p\cdot e^{iu} = 1-p+pe^{iu}.$$

**Example 9.3.2** (Poisson Distribution)**.** If $X \sim \mathcal{P}(\theta)$, then $P(X=k) = e^{-\theta}\frac{\theta^k}{k!}$ for $k \in \mathbb{N}$.

$$\begin{aligned}
\varphi_X(u) = E[e^{iux}] &= \sum_{k=0}^{+\infty} e^{iuk} P(X=k) \\
&= \sum_{k=0}^{+\infty} e^{iuk} e^{-\theta}\frac{\theta^k}{k!} \\
&= e^{-\theta}\sum_{k=0}^{+\infty}\frac{(\theta e^{iu})^k}{k!} \\
&= e^{-\theta}e^{\theta e^{iu}} = e^{\theta(e^{iu}-1)}.
\end{aligned}$$

**Example 9.3.3** (Exponential Distribution)**.** If $X \sim \mathcal{E}(\alpha)$, its density is $f_X(x) = \alpha e^{-\alpha x}$ for $x \geq 0$.

$$\begin{aligned}
\varphi_X(u) = E[e^{iux}] &= \int_0^{+\infty} e^{iux}\alpha e^{-\alpha x}dx \\
&= \alpha\int_0^{+\infty} e^{(iu-\alpha)x}dx \\
&= \alpha\left[\frac{e^{(iu-\alpha)x}}{iu-\alpha}\right]_0^{+\infty} \\
&= \alpha\left(0 - \frac{1}{iu-\alpha}\right) = \frac{\alpha}{\alpha - iu}.
\end{aligned}$$

**Example 9.3.4** (The Gaussian Distribution)**.** Let $X$ be a random variable with a normal distribution $\mathcal{N}(m,\sigma^2)$ with $m \in \mathbb{R}$ and $\sigma > 0$. Let $Z = \frac{X-m}{\sigma}$. Then $Z$ follows a standard normal distribution, $Z \sim \mathcal{N}(0,1)$. First, let us compute $\varphi_Z(u)$.

$$\varphi_Z(u) = E[e^{iuZ}] = \int_{-\infty}^{+\infty} e^{iuz}\frac{1}{\sqrt{2\pi}}e^{-z^2/2}dz.$$

The exponent can be rewritten by completing the square:

$$iuz - \frac{z^2}{2} = -\frac{1}{2}(z^2 - 2iuz) = -\frac{1}{2}((z-iu)^2 - (iu)^2) = -\frac{1}{2}(z-iu)^2 - \frac{u^2}{2}.$$

The integral becomes:

$$\varphi_Z(u) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty} e^{-\frac{1}{2}(z-iu)^2 - \frac{u^2}{2}}dz = e^{-u^2/2}\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{+\infty} e^{-\frac{1}{2}(z-iu)^2}dz.$$

Consider the integral $I(u) = \int_{-\infty}^{+\infty} e^{-\frac{1}{2}(x-iu)^2}dx$. We can show that this integral does not depend on $u$. One method is to use contour integration in the complex plane.

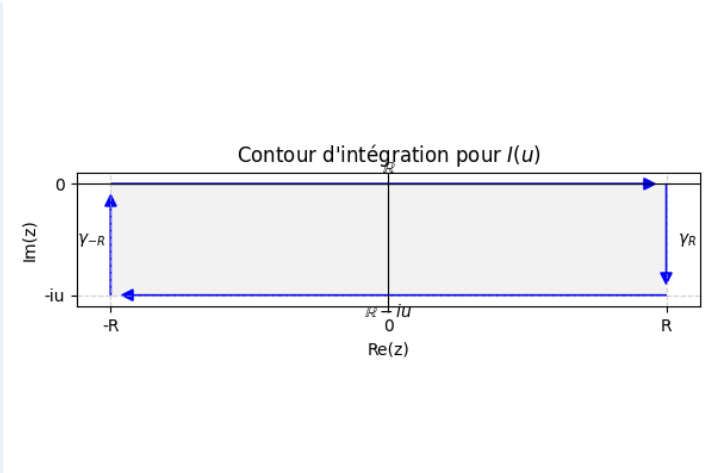



Figure 9.1: The rectangular contour in the complex plane used to show that $I(u)$ is constant.

Another method is to differentiate under the integral sign. Let $f(x,u) = e^{-\frac{1}{2}(x-iu)^2}$. We have $\frac{\partial f}{\partial u} = i(x-iu)e^{-\frac{1}{2}(x-iu)^2}$. This partial derivative is integrable with respect to $x$.

$$\begin{aligned} I'(u) = \int_{\mathbb{R}} \frac{\partial f}{\partial u}(x,u)dx &= \int_{\mathbb{R}} i(x-iu)e^{-\frac{1}{2}(x-iu)^2}dx \\ &= -i\int_{\mathbb{R}} \frac{\partial}{\partial x}\left(e^{-\frac{1}{2}(x-iu)^2}\right)dx \\ &= -i\left[e^{-\frac{1}{2}(x-iu)^2}\right]_{-\infty}^{+\infty} = 0. \end{aligned}$$

Since $I'(u) = 0$, $I(u)$ is a constant. We can evaluate it at $u = 0$:

$$I(u) = I(0) = \int_{-\infty}^{+\infty} e^{-x^2/2}dx = \sqrt{2\pi}.$$

Thus, for $Z \sim \mathcal{N}(0,1)$,

$$\varphi_Z(u) = e^{-u^2/2}\frac{1}{\sqrt{2\pi}}I(u) = e^{-u^2/2}\frac{\sqrt{2\pi}}{\sqrt{2\pi}} = e^{-u^2/2}.$$

Now, for $X \sim \mathcal{N}(m,\sigma^2)$, we have $X = \sigma Z + m$. Using the affine transformation property:

$$\varphi_X(u) = \varphi_{\sigma Z+m}(u) = e^{ium}\varphi_Z(\sigma u) = e^{ium}e^{-(\sigma u)^2/2} = e^{ium-\frac{\sigma^2 u^2}{2}}.$$

We note that the Gaussian density is an eigenvector of the Fourier transform. Let $\psi_\sigma(x) = \frac{1}{\sqrt{2\pi}\sigma}e^{-x^2/(2\sigma^2)}$. We have $\int e^{iux}\psi_\sigma(x)dx = e^{-\sigma^2u^2/2}$. By setting $g_{1/\sigma}(u) = e^{-u^2/(2/\sigma^2)}$, we have $\int e^{iux}\psi_\sigma(x)dx = \psi_{1/\sigma}(u)$.

### 9.3.1 Summary

The following table summarizes the characteristic features for several classical probability distributions.

| **Distribution** $P_X$ | $\varphi_X(u)$ |
|---|---|
| Dirac distribution$\delta_a$ | $e^{iau}$ |
| Bernoulli distribution$Ber(p)$ | $1-p+pe^{iu}$ |
| Poisson distribution$P(\lambda)$ | $\exp(\lambda(e^{iu}-1))$ |
| Geometric distribution$Geo(p)$ | $\frac{pe^{iu}}{1-(1-p)e^{iu}}$ |
| Exponential distribution$Exp(\alpha)$ | $\frac{\alpha}{\alpha-iu}$ |
| Uniform distribution$U(a,b)$ | $\frac{e^{iub}-e^{iua}}{iu(b-a)}$ |
| Normal distribution$N(m,\sigma^2)$ | $\exp\left(ium-\frac{1}{2}\sigma^2u^2\right)$ |

Table 9.1: Characteristic functions of common distributions.

## 9.4 Injectivity Theorem

**Theorem 9.4.1.** Let $\mathcal{M}_1(\mathbb{R})$ be the set of probability measures on $(\mathbb{R},\mathcal{B}(\mathbb{R}))$. The mapping

$$\mu \in \mathcal{M}_1(\mathbb{R}) \mapsto \varphi_\mu$$

is injective.

This fundamental theorem means that the characteristic function uniquely determines the probability law. If two random variables have the same characteristic function, they have the same law.

**Proof** (Proof Idea)**.** The idea is to show that the law of $X$ is determined by the family of laws of $X+\sigma Y$ for $\sigma>0$, where $Y\sim\mathcal{N}(0,1)$ is independent of $X$. The law of $X+\sigma Y$ is a "regularized" (convolved) version of the law of $X$. Let $(\Omega,\mathcal{F},P)$ be a probability space on which two independent random variables are defined: $X\sim\mu$ and $Y\sim\mathcal{N}(0,1)$. For $\sigma>0$, we study the variable $X_\sigma=X+\sigma Y$. The characteristic function of $X_\sigma$ is:

$$\varphi_{X_\sigma}(t)=E[e^{it(X+\sigma Y)}]=E[e^{itX}]E[e^{it\sigma Y}]=\varphi_X(t)\varphi_{\sigma Y}(t)=\varphi_\mu(t)e^{-\frac{\sigma^2t^2}{2}}.$$

This function is determined by $\varphi_\mu(t)$. Let $f:\mathbb{R}\to\mathbb{R}$ be a continuous function with compact support. The expectation $E[f(X+\sigma Y)]$ is given by:

$$\begin{aligned}E[f(X+\sigma Y)]&=\iint_{\mathbb{R}^2}f(x+\sigma y)d\mu(x)dP_Y(y)\\&=\iint f(x+z)d\mu(x)g_\sigma(z)dz,\end{aligned}$$

where $g_\sigma(z)$ is the density of $\sigma Y\sim\mathcal{N}(0,\sigma^2)$. Using the inverse Fourier representation for $f(z)=\frac{1}{2\pi}\int e^{-izt}\hat{f}(t)dt$, where $\hat{f}(t)=\int e^{izt}f(z)dz$ (Fourier transform convention), we can show that $E[f(X+\sigma Y)]$ is completely determined by $\varphi_\mu$. A more direct derivation (as seen in the notes) is as follows:

$$\begin{aligned}E[f(X+\sigma Y)]&=\iint f(x+y')d\mu(x)g_\sigma(y')dy'\\\text{change of variable } z&=x+y'\\&=\iint f(z)g_\sigma(z-x)d\mu(x)dz\\&=\int f(z)\left(\int g_\sigma(z-x)d\mu(x)\right)dz.\end{aligned}$$

The inner integral is the density of the law of $X+\sigma Y$, which is the convolution of the measure $\mu$ and the Gaussian density $g_\sigma$. Since the Fourier transform of a convolution is the product of the Fourier

transforms, the law of $X + \sigma Y$ has the characteristic function $\varphi_\mu(t)e^{-\sigma^2 t^2/2}$. Since the characteristic function determines the law, the law of $X + \sigma Y$ is uniquely determined by $\varphi_\mu$. Now, let $\sigma \to 0$. The random variable $X + \sigma Y$ converges in probability to $X$. Indeed, for all $\epsilon > 0$:

$$P(|(X + \sigma Y) - X| > \epsilon) = P(|\sigma Y| > \epsilon) = P\left(|Y| > \frac{\epsilon}{\sigma}\right).$$

When $\sigma \to 0$, $\frac{\epsilon}{\sigma} \to +\infty$. Therefore,

$$\lim_{\sigma \to 0} P\left(|Y| > \frac{\epsilon}{\sigma}\right) = P(|Y| = \infty) = 0.$$

Thus, $X + \sigma Y \xrightarrow{P} X$. Convergence in probability implies convergence in law: $X + \sigma Y \xrightarrow{d} X$. Since the law of $X + \sigma Y$ is determined by $\varphi_\mu$ for all $\sigma > 0$, its limit in law, which is the law of $X$, is also determined by $\varphi_\mu$. This proves injectivity. The convergence in law ($X_n \xrightarrow{d} X$) is equivalent to the convergence of the distribution functions $F_{X_n}(x) \to F_X(x)$ for all points $x$ where $F_X$ is continuous. This can be shown by the following inequalities. Let $x$ be a continuity point of $F_X$. For all $\epsilon > 0$:

$$P(X_n \leq x) \leq P(|X_n - X| \geq \epsilon) + P(X \leq x + \epsilon).$$

By taking the limit supremum as $n \to \infty$, and knowing that $X_n \xrightarrow{P} X$ (which implies $P(|X_n - X| \geq \epsilon) \to 0$), we obtain

$$\limsup_{n \to \infty} F_{X_n}(x) \leq F_X(x + \epsilon).$$

Similarly,

$$P(X \leq x - \epsilon) \leq P(|X_n - X| \geq \epsilon) + P(X_n \leq x).$$

By taking the limit infimum, we obtain

$$F_X(x - \epsilon) \leq \liminf_{n \to \infty} F_{X_n}(x).$$

Combining these two, for a continuity point $x$ of $F_X$, we can let $\epsilon \to 0$:

$$F_X(x) \leq \liminf_{n \to \infty} F_{X_n}(x) \leq \limsup_{n \to \infty} F_{X_n}(x) \leq F_X(x),$$

which proves that $\lim_{n \to \infty} F_{X_n}(x) = F_X(x)$. $\square$

# CHAPTER 10

# CENTRAL LIMIT THEOREM

## 10.1 Paul Lévy's theorem

**Theorem 10.1.1** (Paul Lévy)**.** Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of real random variables and let $X$ be a real random variable. Let $\phi_{X_n}$ and $\phi_X$ be their characteristic functions. The following are equivalent:

i) $X_n \xrightarrow{\mathcal{L}} X$

ii) $\forall u \in \mathbb{R}, \lim_{n\to\infty} \phi_{X_n}(u) = \phi_X(u)$

**Proof.** Direct implication: i) $\implies$ ii)
We have $\phi_{X_n}(u) = \mathbb{E}[e^{iuX_n}] = \mathbb{E}[\cos(uX_n)] + i\mathbb{E}[\sin(uX_n)]$. The functions $x \mapsto \cos(ux)$ and $x \mapsto \sin(ux)$ (with $u$ fixed) are continuous and bounded. By the definition of convergence in distribution,

$$\mathbb{E}[\cos(uX_n)] \xrightarrow[n\to\infty]{} \mathbb{E}[\cos(uX)]$$
$$\mathbb{E}[\sin(uX_n)] \xrightarrow[n\to\infty]{} \mathbb{E}[\sin(uX)]$$

and thus $\phi_{X_n}(u) \xrightarrow[n\to\infty]{} \phi_X(u)$. Converse implication: ii) $\implies$ i)

We use the same strategy as for proving the injectivity of the Fourier transform of probability measures. We assume that $X, X_n$ for $n \in \mathbb{N}$ are all defined on the same probability space $(\Omega, \mathcal{F}, P)$, on which a random variable $Y$ is also defined, independent of $X, X_n$ for $n \in \mathbb{N}$, with distribution $\mathcal{N}(0,1)$. Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function with compact support. We assume ii) holds, and we want to show that $X_n \xrightarrow{\mathcal{L}} X$. Let $\sigma > 0$ and write:

$$\begin{aligned}\mathbb{E}[f(X_n)] - \mathbb{E}[f(X)] = \mathbb{E}[f(X_n) - f(X_n + \sigma Y)] + \mathbb{E}[f(X_n + \sigma Y)] - \mathbb{E}[f(X + \sigma Y)] \\ + \mathbb{E}[f(X + \sigma Y) - f(X)]\end{aligned}$$

Since $f$ is continuous with compact support, it is uniformly continuous.

$$\forall \varepsilon > 0, \exists \delta > 0, \forall x, y \in \mathbb{R}, |x - y| < \delta \implies |f(x) - f(y)| < \varepsilon$$

Let $\varepsilon > 0$ be fixed. Let $\delta$ be associated with $\varepsilon$ as above. We write:

$$\begin{aligned}|\mathbb{E}[f(X_n) - f(X_n + \sigma Y)]| &\leq \mathbb{E}[|f(X_n) - f(X_n + \sigma Y)|] \\ &= \int_{|\sigma Y| \leq \delta} |f(X_n) - f(X_n + \sigma Y)| dP + \int_{|\sigma Y| > \delta} |f(X_n) - f(X_n + \sigma Y)| dP\end{aligned}$$

$$\int_{|\sigma Y| \leq \delta} |f(X_n) - f(X_n + \sigma Y)| dP < \varepsilon$$

$$\int_{|\sigma Y| > \delta} |f(X_n) - f(X_n + \sigma Y)| dP \leq \int_{|\sigma Y| > \delta} 2\|f\|_\infty dP = 2\|f\|_\infty \mathbb{P}(|\sigma Y| > \delta)$$

We have $\lim_{\sigma \to 0} \mathbb{P}(|\sigma Y| > \delta) = \lim_{\sigma \to 0} \mathbb{P}(|Y| > \delta/\sigma) = 0$. Thus, $\exists \sigma_0$ such that for $\sigma < \sigma_0$, $2\|f\|_\infty \mathbb{P}(|\sigma Y| > \delta) < \varepsilon$. We thus have $|\mathbb{E}[f(X_n) - f(X_n + \sigma Y)]| < 2\varepsilon$. By the same argument, we have $|\mathbb{E}[f(X + \sigma Y) - f(X)]| < 2\varepsilon$. Now consider the middle term.

$$\forall n \in \mathbb{N}, \quad |\mathbb{E}[f(X_n)] - \mathbb{E}[f(X)]| \leq 2\varepsilon + |\mathbb{E}[f(X_n + \sigma Y) - f(X + \sigma Y)]| + 2\varepsilon$$

We use the formula from the previous lecture (via convolution with a Gaussian):

$$\mathbb{E}[f(X_n + \sigma Y)] = \iint f(z) e^{iuz} \frac{1}{\sqrt{2\pi}\sigma} g_{1/\sigma}(u) \phi_{X_n}(-u) du dz$$

$$\mathbb{E}[f(X + \sigma Y)] = \iint f(z) e^{iuz} \frac{1}{\sqrt{2\pi}\sigma} g_{1/\sigma}(u) \phi_{X}(-u) du dz$$

where $g_{1/\sigma}$ is the density of $\mathcal{N}(0, 1/\sigma^2)$. Set $F_n(u, z) = f(z) e^{iuz} \frac{1}{\sqrt{2\pi}\sigma} g_{1/\sigma}(u) \phi_{X_n}(-u)$. We have:

$$|F_n(u, z)| \leq |f(z)| \frac{1}{\sqrt{2\pi}\sigma} g_{1/\sigma}(u)$$

This bound does not depend on $n$ and is in $L^1(d\lambda(u), d\lambda(z)) = L^1(\mathbb{R}^2, \lambda^{\otimes 2})$. By the dominated convergence theorem (DCT), since $\phi_{X_n}(u) \to \phi_X(u)$, we have:

$$\exists N, \forall n \geq N, \quad |\mathbb{E}[f(X_n + \sigma Y)] - \mathbb{E}[f(X + \sigma Y)]| < \varepsilon$$

Then, for $n \geq N$,

$$|\mathbb{E}[f(X_n)] - \mathbb{E}[f(X)]| < 2\varepsilon + \varepsilon + 2\varepsilon = 5\varepsilon$$

Since this holds for any $\varepsilon > 0$, we have $\mathbb{E}[f(X_n)] \to \mathbb{E}[f(X)]$ for any continuous function $f$ with compact support, which is the definition of convergence in distribution. □

## 10.2 Moments

**Proposition 10.2.1.** Let $X$ be a real-valued random variable. Let $k \geq 1$. Suppose that $\mathbb{E}[|X|^k] < \infty$. Then, $\phi_X$ is differentiable $k$ times, and furthermore,

$$\phi_X^{(k)}(0) = i^k \mathbb{E}[X^k]$$

**Proof.** Let's prove the case for $k = 1$. We have $\mathbb{E}[|X|] < \infty$. The characteristic function is $\phi_X(u) = \int_{\mathbb{R}} e^{iux} dP_X(x)$. This is a parameterized integral, and we want to derive with respect to $u$. Let $F(x, u) = e^{iux}$. We have $\frac{\partial F}{\partial u}(x, u) = ixe^{iux}$. We have $\left|\frac{\partial F}{\partial u}(x, u)\right| = |ixe^{iux}| = |x| =$ —x—. The dominating function $|x|$ is $P_X$-integrable because $\mathbb{E}[|X|] < \infty$, and it does not depend on $u$. By the theorem of

differentiation under the integral sign, $\phi_X$ is differentiable and

$$\phi'_X(u) = \int_{\mathbb{R}} \frac{\partial F}{\partial u}(x,u)dP_X(x) = \int_{\mathbb{R}} ixe^{iux}dP_X(x)$$

Thus $\phi'_X(0) = \int_{\mathbb{R}} ixdP_X(x) = i\mathbb{E}[X]$. The result for $k > 1$ is obtained by induction. □

This result makes it easy to find the sequence of moments.

### Examples

**Example 10.2.2** (Uniform Distribution)**.** Let $X \sim \mathcal{U}([-1,1])$. The density is $f(x) = \frac{1}{2}\mathbf{1}_{[-1,1]}(x)$.

$$\phi_X(u) = \int_{-1}^{1} e^{iux}\frac{1}{2}dx = \frac{1}{2}\left[\frac{e^{iux}}{iu}\right]_{-1}^{1} = \frac{e^{iu} - e^{-iu}}{2iu} = \frac{\sin(u)}{u}$$

We know the series expansion of $\sin(u) = \sum_{n=0}^{\infty}(-1)^n \frac{u^{2n+1}}{(2n+1)!}$. Thus, $\phi_X(u) = \frac{\sin(u)}{u} = \sum_{n=0}^{\infty}(-1)^n \frac{u^{2n}}{(2n+1)!}$. The derivative of order $2n$ at 0 is $(2n)!$ times the coefficient of $u^{2n}$:

$$\phi_X^{(2n)}(0) = (2n)!\frac{(-1)^n}{(2n+1)!}$$

Hence, $\mathbb{E}[X^{2n}] = \frac{\phi_X^{(2n)}(0)}{i^{2n}} = \frac{1}{(-1)^n}(2n)!\frac{(-1)^n}{(2n+1)!} = \frac{1}{2n+1}$. Easier: $\mathbb{E}[X^{2n}] = \int_{-1}^{1} x^{2n}\frac{1}{2}dx = \frac{1}{2}\left[\frac{x^{2n+1}}{2n+1}\right]_{-1}^{1} = \frac{1}{2n+1}$.

**Example 10.2.3** (Gaussian Distribution)**.** Let $X \sim \mathcal{N}(0,1)$. We know that $\phi_X(u) = e^{-u^2/2}$. The series expansion is:

$$\phi_X(u) = e^{-u^2/2} = \sum_{n=0}^{\infty}\frac{1}{n!}\left(-\frac{u^2}{2}\right)^n = \sum_{n=0}^{\infty}\frac{(-1)^n}{n!2^n}u^{2n}$$

We deduce that $\phi_X^{(2n)}(0) = (2n)!\frac{(-1)^n}{n!2^n}$. Thus, $\mathbb{E}[X^{2n}] = \frac{\phi_X^{(2n)}(0)}{i^{2n}} = \frac{1}{(-1)^n}\frac{(2n)!(-1)^n}{n!2^n} = \frac{(2n)!}{n!2^n}$.

## 10.3 Convolution

**Lemma 10.3.1.** Let $X, Y$ be two independent random variables. Then $\forall u \in \mathbb{R}$,

$$\phi_{X+Y}(u) = \phi_X(u)\phi_Y(u)$$

**Proof.** $\phi_{X+Y}(u) = \mathbb{E}[e^{iu(X+Y)}] = \mathbb{E}[e^{iuX}e^{iuY}]$. By the independence of $X$ and $Y$, the r.v. $e^{iuX}$ and $e^{iuY}$ are also independent. Thus,

$$\mathbb{E}[e^{iuX}e^{iuY}] = \mathbb{E}[e^{iuX}]\mathbb{E}[e^{iuY}] = \phi_X(u)\phi_Y(u)$$

□

**Corollary 10.3.2.** If $X_1, \dots, X_n$ are $n$ independent random variables, then $\phi_{X_1+\dots+X_n}(u) = \prod_{i=1}^n \phi_{X_i}(u)$. If, furthermore, they are identically distributed, $\phi_{X_1+\dots+X_n}(u) = (\phi_{X_1}(u))^n$.

Suppose that $X$ has a density law $f$ and $Y$ has a density law $g$, and $X, Y$ are independent.

$$\phi_{X+Y}(u) = \iint_{\mathbb{R}^2} e^{iu(x+y)} f(x)g(y)dxdy$$

We set $z = x + y$. We keep $x$ and replace $y = z - x$.

$$\begin{aligned}\phi_{X+Y}(u) &= \iint e^{iuz} f(x)g(z-x)dxdz \quad \text{(Fubini)} \\ &= \int e^{iuz} \left( \int f(x)g(z-x)dx \right) dz = \int e^{iuz} h(z)dz\end{aligned}$$

where $h(z) = \int f(x)g(z-x)dx = (f * g)(z)$ is the convolution product of $f$ and $g$. By identifying these two formulas, we see that $X + Y$ has a density function $f * g$. Whenever $f$ and $g$ are in $L^1$, the convolution product $f * g$ is well defined and also in $L^1$.

## 10.4 Central Limit Theorem

**Theorem 10.4.1** (Central Limit Theorem)**.** Let $(X_n)_{n\geq 1}$ be a sequence of i.i.d. random variables (independent and identically distributed) with a second-order moment. Let $m = \mathbb{E}[X_i]$ and $\sigma^2 = \mathrm{Var}[X_i]$. Then,

$$\frac{X_1 + \dots + X_n - nm}{\sqrt{n}} \xrightarrow[n\to\infty]{\mathcal{L}} \mathcal{N}(0, \sigma^2)$$

**Warning:** It is absolutely necessary to center the random variables by subtracting $\mathbb{E}[X_1 + \dots + X_n] = nm$. However, we can normalize to obtain a variance of 1:

$$\frac{X_1 + \dots + X_n - nm}{\sqrt{n}\sigma} \xrightarrow[n\to\infty]{\mathcal{L}} \mathcal{N}(0, 1)$$

### History

This is a universal result. As soon as the random variables $(X_n)$ have a second-order moment, the fluctuations of $X_1, \dots, X_n$ around their mean are described by the Gaussian law. The CLT (or CLT) is a refinement of the law of large numbers, assuming the existence of a second-order moment.

- **Abraham de Moivre (1733)**: *The doctrine of chances.*
- **Pierre Simon de Laplace (1812)**: *Traité de Mécanique céleste* and *Théorie analytique des probabilités* (CLT for the binomial law, Moivre-Laplace theorem).
- **Lindeberg**, **Turing (1934)**.

**Proof** (Preuve du TCL)**.** Let $Y_n = \frac{X_1+\dots+X_n-nm}{\sqrt{n}}$ and $\tilde{X}_k = X_k - m$ for $1 \leq k \leq n$. We have $\mathbb{E}[\tilde{X}_k] = 0$ and $\mathbb{E}[\tilde{X}_k^2] = \sigma^2$. We can rewrite $Y_n = \frac{1}{\sqrt{n}} \sum_{k=1}^n \tilde{X}_k$. Let us study the convergence in distribution of

$Y_n$ using characteristic functions. The $\tilde{X}_k$ are i.i.d.

$$\begin{aligned}\phi_{Y_n}(u) = \mathbb{E}\left[e^{iuY_n}\right] = \mathbb{E}\left[e^{i\frac{u}{\sqrt{n}}\sum_{k=1}^n \tilde{X}_k}\right] &= \mathbb{E}\left[\prod_{k=1}^n e^{i\frac{u}{\sqrt{n}}\tilde{X}_k}\right] \\ &= \prod_{k=1}^n \mathbb{E}\left[e^{i\frac{u}{\sqrt{n}}\tilde{X}_k}\right] \quad \text{(by independence)} \\ &= \left(\phi_{\tilde{X}_1}\left(\frac{u}{\sqrt{n}}\right)\right)^n \quad \text{(since i.i.d.)}\end{aligned}$$

Since $\mathbb{E}[\tilde{X}_1^2] < \infty$, $\phi_{\tilde{X}_1}$ is twice differentiable. It therefore admits a second-order Taylor expansion at 0:

$$\phi_{\tilde{X}_1}(v) = \phi_{\tilde{X}_1}(0) + v\phi'_{\tilde{X}_1}(0) + \frac{v^2}{2}\phi''_{\tilde{X}_1}(0) + o(v^2)$$

Let us compute the coefficients:

- $\phi_{\tilde{X}_1}(0) = \mathbb{E}[e^0] = 1$
- $\phi'_{\tilde{X}_1}(0) = i\mathbb{E}[\tilde{X}_1] = i \cdot 0 = 0$
- $\phi''_{\tilde{X}_1}(0) = i^2\mathbb{E}[\tilde{X}_1^2] = -\sigma^2$

Thus, $\phi_{\tilde{X}_1}(v) = 1 - \frac{\sigma^2 v^2}{2} + o(v^2)$. Applying this expansion with $v = u/\sqrt{n}$:

$$\phi_{\tilde{X}_1}\left(\frac{u}{\sqrt{n}}\right) = 1 - \frac{\sigma^2}{2}\left(\frac{u}{\sqrt{n}}\right)^2 + o\left(\left(\frac{u}{\sqrt{n}}\right)^2\right) = 1 - \frac{\sigma^2 u^2}{2n} + o\left(\frac{1}{n}\right)$$

Now, we raise to the power of $n$:

$$\phi_{Y_n}(u) = \left(1 - \frac{\sigma^2 u^2}{2n} + o\left(\frac{1}{n}\right)\right)^n$$

To find the limit, we use the (complex) logarithm. We know that as $z \to 0$, $\ln(1+z) = z + O(z^2)$.

$$\begin{aligned}\ln(\phi_{Y_n}(u)) &= n\ln\left(1 - \frac{\sigma^2 u^2}{2n} + o\left(\frac{1}{n}\right)\right) \\ &= n\left(-\frac{\sigma^2 u^2}{2n} + o\left(\frac{1}{n}\right)\right) \\ &= -\frac{\sigma^2 u^2}{2} + n \cdot o\left(\frac{1}{n}\right) = -\frac{\sigma^2 u^2}{2} + o(1)\end{aligned}$$

Therefore, $\lim_{n\to\infty} \ln(\phi_{Y_n}(u)) = -\frac{\sigma^2 u^2}{2}$. By continuity of the exponential,

$$\lim_{n\to\infty} \phi_{Y_n}(u) = e^{-\frac{\sigma^2 u^2}{2}}$$

We recognize the characteristic function of the $\mathcal{N}(0, \sigma^2)$ distribution. We conclude by Paul Lévy's theorem that $Y_n \xrightarrow{\mathcal{L}} \mathcal{N}(0, \sigma^2)$. $\square$

# CHAPTER 11

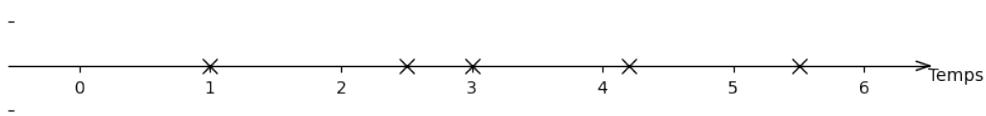
# POISSON PROCESS

## 11.1 Introduction and Motivation

We want to model events that occur at precise but unpredictable dates.

> **Example 11.1.1.** Examples of such events include earthquakes, atomic decay, rain, or the beginning and end of telephone calls.

At each occurrence of an event, we mark an impact on the time axis.




## 11.2 Model hypotheses

To construct our model, we make the following assumptions:

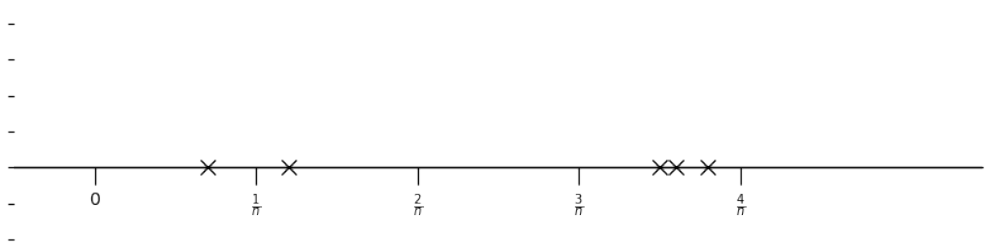
i) The experimental conditions do not change over time.

ii) The number of impacts in disjoint time intervals is independent.

For $t \geq 0$, we denote $N(t)$ the number of occurrences in the interval $[0, t]$. We thus have a family of random variables $(N(t))_{t \geq 0}$ indexed by a continuous parameter. This is a **stochastic process**.

## 11.3 Construction of a discrete model

Let $n \geq 1$ be an integer. We subdivide the real axis into intervals of length $\frac{1}{n}$, of the form $\left[\frac{k}{n}, \frac{k+1}{n}\right[$, for $k \geq 0$.



A range can have $0, 1, 2, \ldots$ impacts.

**Definition 11.3.1.** An interval $\left[\frac{k}{n}, \frac{k+1}{n}\right[$ is called **occupied** if it contains at least one hit, and **empty** otherwise.

Let $p_n$ denote the probability that an interval $\left[\frac{k}{n}, \frac{k+1}{n}\right[$ is occupied.
According to hypothesis (i), $p_n$ does not depend on the interval.
For $t > 0$, let $N_n(t)$ be the number of intervals $\left[\frac{k}{n}, \frac{k+1}{n}\right[$ (with $\frac{k+1}{n} \leq t$) that are occupied.
By hypothesis (ii), the states (empty or occupied) of these intervals are independent, and $N_n(t)$ follows the binomial distribution $B(\lfloor nt \rfloor, p_n)$.

## 11.4 Convergence to the Poisson distribution

We study the limit in distribution of $N_n(t)$, for example using characteristic functions. The characteristic function of $N_n(t)$ is:

$$\varphi_{N_n(t)}(u) = E[e^{iuN_n(t)}] = (1 - p_n + p_n e^{iu})^{\lfloor nt \rfloor}$$

We add an additional assumption: $p_n \to 0$ when $n \to \infty$. Otherwise, the model "explodes". To study the limit, we analyze the logarithm of the characteristic function:

$$\begin{aligned} \ln(\varphi_{N_n(t)}(u)) &= \lfloor nt \rfloor \ln(1 - p_n + p_n e^{iu}) \\ &= \lfloor nt \rfloor \ln(1 + p_n(e^{iu} - 1)) \end{aligned}$$

By using the limited expansion $\ln(1 + x) = x + o(x)$ for $x \to 0$, we obtain:

$$\begin{aligned} \ln(\varphi_{N_n(t)}(u)) &= \lfloor nt \rfloor \left(p_n(e^{iu} - 1) + o(p_n)\right) \\ &= \lfloor nt \rfloor p_n(e^{iu} - 1) + o(np_n) \end{aligned}$$

By transitioning to the exponential form:

$$\varphi_{N_n(t)}(u) = e^{\lfloor nt \rfloor p_n(e^{iu} - 1) + o(np_n)}$$

Assume that $np_n \to \lambda$ as $n \to \infty$, where $\lambda$ is a positive constant. Then $\lfloor nt \rfloor p_n \approx ntp_n \to \lambda t$. The limit of the characteristic function becomes:

$$\lim_{n \to \infty} \varphi_{N_n(t)}(u) = e^{\lambda t(e^{iu} - 1)}$$

We recognize the characteristic function of a Poisson distribution with parameter $\lambda t$. Thus, $N_n(t)$ converges in law to a random variable following a Poisson distribution $\mathcal{P}(\lambda t)$.

$$N_n(t) \xrightarrow{\mathcal{L}} \mathcal{P}(\lambda t)$$

**Remark 11.4.1** (Reminder on the Poisson Distribution)**.** A random variable $X$ taking values in $\mathbb{N}$ follows a Poisson distribution with parameter $\lambda > 0$, denoted $X \sim \mathcal{P}(\lambda)$, if for all $k \in \mathbb{N}$:

$$P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}$$

Its **expectation** and its **variance** are equal to $\lambda$, i.e., $E[X] = \mathrm{Var}[X] = \lambda$.
Its **characteristic function** is $\varphi_X(u) = \exp(\lambda(e^{iu} - 1))$.

## 11.5 Jump Instants

We have shown that, at fixed $t$, $N_n(t) \xrightarrow{\mathcal{L}} \mathcal{P}(\lambda t)$. However, the function $t \mapsto N_n(t)$ is a **random step function**, which is zero at 0, increasing, and whose increments are either 0 or 1.

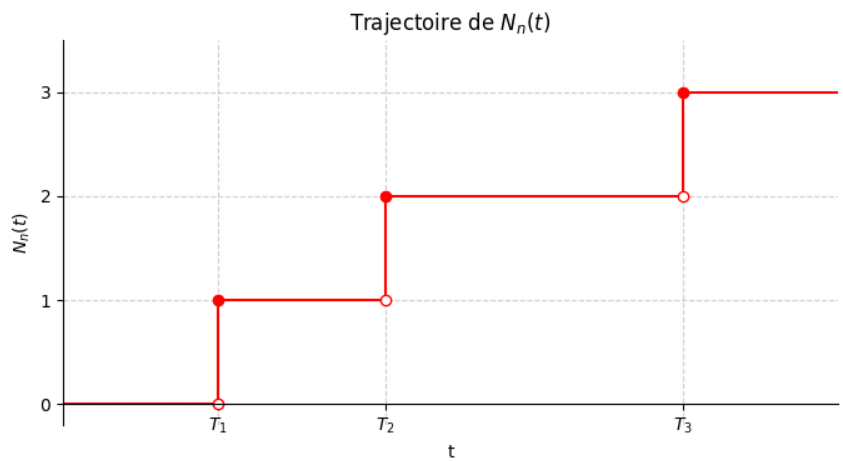



Figure 11.1: A realization of the random function $t \mapsto N_n(t)$.

The question arises: does the random function $t \mapsto N_n(t)$ converge when $n \to \infty$? The function $N_n(t)$ is completely determined by its jump times. We define the jump times of $N_n(t)$ as:

$$\forall k \geq 1, \quad T_k^n = \inf\{t \geq 0 : N_n(t) = k\}$$

Indeed, we have $N_n(t) = \sum_{k=1}^{\infty} 1_{\{T_k^n \leq t\}}$. Let us examine the convergence of these jump times, starting with $T_1^n$. Let $i \geq 0$ be an integer. The event $\{T_1^n > i/n\}$ means that the first $i$ time intervals, $[0, 1/n), [1/n, 2/n), \ldots, [(i-1)/n, i/n),$, are all empty. To analyze this, we introduce a sequence of random variables $X_k$ for $k \geq 0$:

$$X_k = \begin{cases} 0 & \text{si l'intervalle } [k/n, (k+1)/n[ \text{ est vide} \\ 1 & \text{sinon (occupé)} \end{cases}$$

Then $P(T_1^n > i/n) = P(X_0 = 0, \ldots, X_{i-1} = 0)$. By independence, this equals $(1 - p_n)^i$. For a fixed $t > 0$, let us choose $i = \lfloor nt \rfloor$. Then:

$$P(T_1^n > t) \approx P(T_1^n > \lfloor nt \rfloor / n) = (1 - p_n)^{\lfloor nt \rfloor}$$

Since we assumed $np_n \to \lambda$, we have $p_n \approx \lambda/n$.

$$P(T_1^n > t) \approx \left(1 - \frac{\lambda}{n}\right)^{nt} \xrightarrow[n \to \infty]{} e^{-\lambda t}$$

We recognize the survival function of an exponential distribution. Therefore, $T_1^n \xrightarrow{\mathcal{L}} \text{Exp}(\lambda)$.

**Remark 11.5.1** (Reminder on the Exponential Distribution). A random variable $X$ follows the exponential distribution with parameter $\lambda > 0$, denoted $\text{Exp}(\lambda)$, if its density function is $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$. Its expectation is $E[X] = 1/\lambda$ and its variance is $\text{Var}[X] = 1/\lambda^2$. Its cumulative distribution function is $F_X(x) = P(X \leq x) = 1 - e^{-\lambda x}$ for $x \geq 0$.

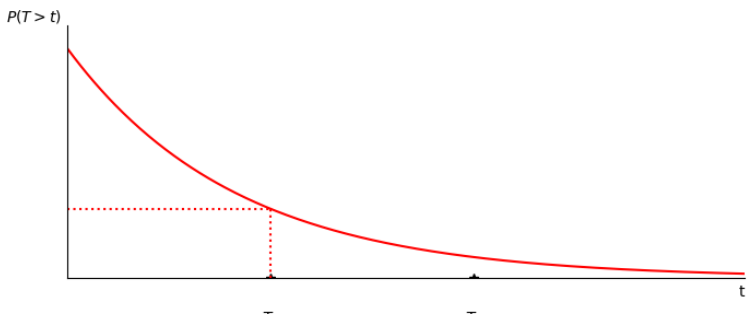



Figure 11.2: Survival function of the exponential distribution.

Furthermore, it can be shown that the durations between successive jumps are independent and identically distributed. For example, $T_2^n - T_1^n$ is independent of $T_1^n$ and also converges in distribution to $\text{Exp}(\lambda)$. Let us verify this for the discrete model.

**Remark 11.5.2.** Let $i, j \geq 1$.

$$P(T_1^n = i/n, T_2^n - T_1^n = j/n) = P(X_0 = 0, \ldots, X_{i-1} = 0, X_i = 1, X_{i+1} = 0, \ldots, X_{i+j-1} = 0, X_{i+j} = 1)$$

. This probability equals $(1-p_n)^{i-1}p_n \cdot (1-p_n)^{j-1}p_n$. This is equal to $P(T_1^n = i/n) \cdot P(T_1^n = j/n)$. Hence the result.
Thus, the inter-jump vector converges in distribution:

$$(T_1^n, T_2^n - T_1^n, \ldots, T_k^n - T_{k-1}^n) \xrightarrow[n\to\infty]{\mathcal{L}} (\text{Exp}(\lambda))^{\otimes k}$$

that is, toward a vector of $k$ independent and identically distributed random variables with distribution $\text{Exp}(\lambda)$.

## 11.6 The Continuous Poisson Process

The previous calculations allow us to envision the candidate for the limit in law of $N_n(t)$.

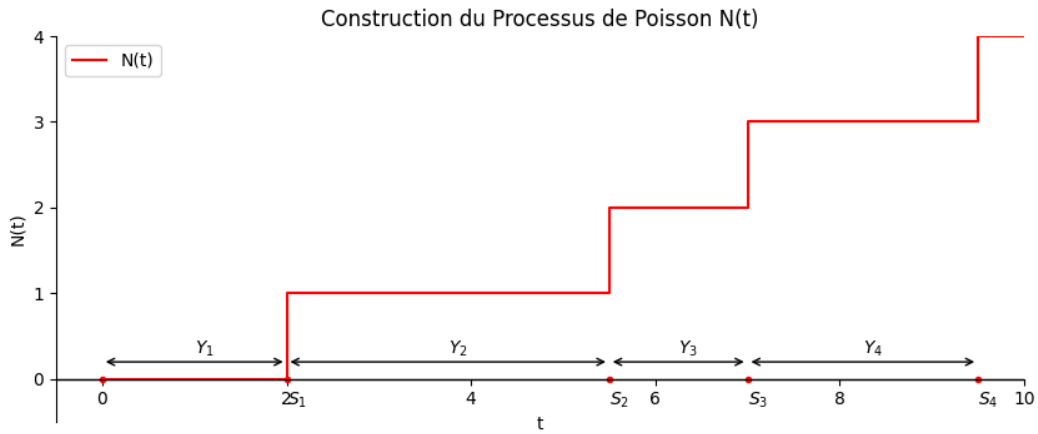
### 11.6.1 Definition from jump moments

Let $(Y_n)_{n\geq 1}$ be a sequence of independent and identically distributed random variables following the exponential distribution $\text{Exp}(\lambda)$. Let $S_0 = 0$ and for $n \geq 1$, $S_n = Y_1 + Y_2 + \cdots + Y_n$. The variable $S_n$ represents the time of the $n$-th event.

**Definition 11.6.1** (Poisson Process)**.** For any $t \geq 0$, we define the number of events up to time $t$ as:

$$N(t) = \max\{n \geq 0 : S_n \leq t\}$$

The stochastic process $(N(t))_{t\geq 0}$ is called the **Poisson process** with intensity $\lambda$.




For each $t$, $N(t)$ is a random variable that follows a Poisson distribution $\mathcal{P}(\lambda t)$. We have thus constructed a continuum of random variables $(N(t))_{t\geq 0}$ on the same probability space. We have $S_n \to +\infty$ almost surely. $N(t)$ is the unique index $k$ such that $S_k \leq t < S_{k+1}$.

## 11.7 Applications

### 11.7.1 Application 1: Bombardment of London (WWII)

During World War II, the city of London suffered 537 bomb impacts. The map of the city was divided into a grid of 576 squares of 0.25 km$^2$ each. The average number of impacts per square is therefore $\lambda = 537/576 \approx 0.9323$. We compare the number of squares receiving $k$ impacts with the predictions of the Poisson distribution $\mathcal{P}(0.9323)$.

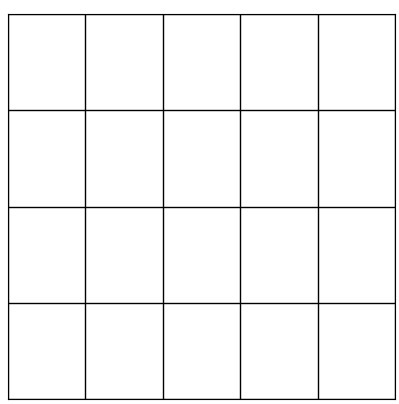


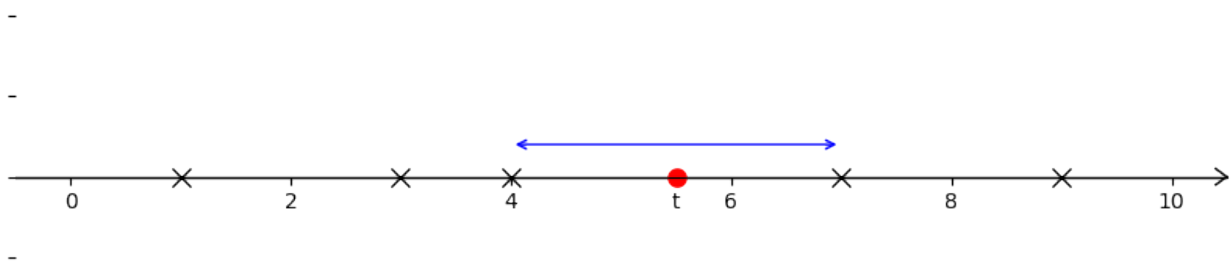
Figure 11.3: Grid of the London map.

The following table compares the observed data (number of squares with $k$ impacts) and the expected values according to the Poisson distribution ($576 \times P(X = k)$ with $X \sim \mathcal{P}(0.9323)$).

| **k (impacts)** | **0** | **1** | **2** | **3** | **4** | $\geq 5$ |
|---|---|---|---|---|---|---|
| **Observed** | 229 | 211 | 93 | 35 | 7 | 1 |
| **Expected** | 226.74 | 211.39 | 98.54 | 30.62 | 7.14 | 1.57 |

### 11.7.2 Application 2: Bus Problem

Bus arrivals at a stop are modeled by a Poisson process with intensity $\lambda$. The average time between two arrivals is thus $1/\lambda$. I arrive at time $t$. What will my waiting time be?




- **First reasoning:** Due to the memoryless property of the exponential distribution governing the inter-arrival times, my waiting time follows an $\text{Exp}(\lambda)$ distribution, with an average of $1/\lambda$.
- **Second reasoning:** I arrive randomly between two buses. By symmetry, the average waiting time would be half the typical interval length, i.e., $\frac{1}{2\lambda}$. This reasoning is incorrect.

Let $S_n = Y_1 + \cdots + Y_n$ be the arrival times of the buses. Let $k$ be such that $S_{k-1} < t \leq S_k$. My waiting time is $S_k - t$. The error comes from the fact that the interval $[S_{k-1}, S_k]$ in which we "fall" is not a typical interval. There is a higher chance of falling into a large interval. Typically, this interval is twice as large as the average. The interval $Y_k = S_k - S_{k-1}$ does not follow the law $\text{Exp}(\lambda)$ because its distribution depends on $t$.